Name [redacted] **ID#** [redacted]

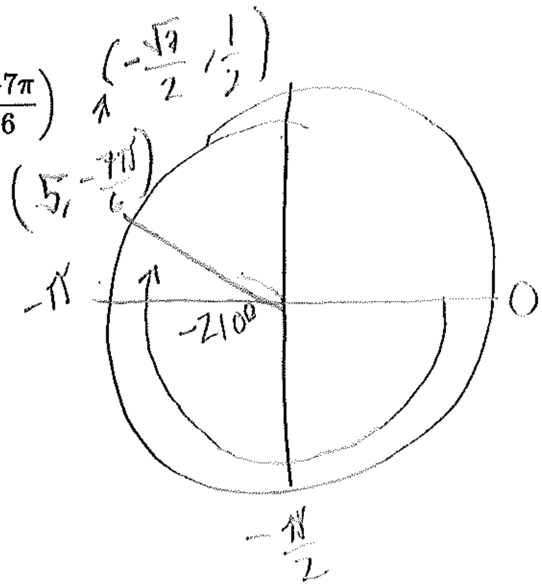
Math 104 Spring 2025 Final Exam

- A non graphing calculator is allowed
- Must show all work to receive credit
- Written or printed notes allowed (notes on digital devices not allowed)
- Attach all scratch paper

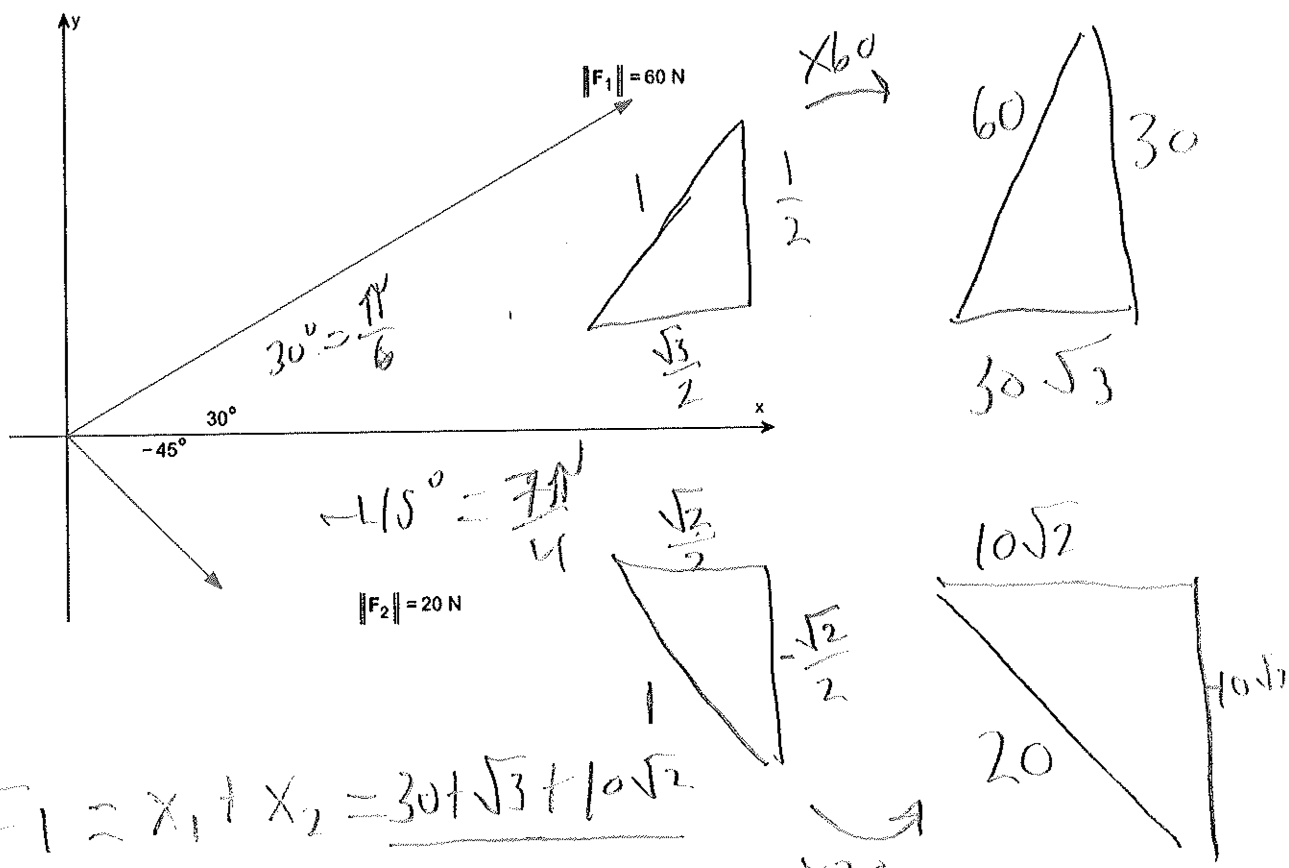
Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

SSA or SAS

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$



$F_1 = x_1 + x_2 = \underline{30+\sqrt{3}+10\sqrt{2}}$

$F_2 = y_1 + y_2 = 30 - 10\sqrt{2}$

$F_1 + F_2 = (30+\sqrt{3}+10\sqrt{2}) + (30-10\sqrt{2})$

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2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

rate of change formula $= \dfrac{f(b) - fa}{b - a}$

$4 - 1^2 = 1$

$a = -1$ $4 - (-1)^2 = 3$

$-5 - 3 = -8$

$3 + 1 = 4$

$= \boxed{-2}$

$(3,3)$

$4 - (3)^2 = -5$

$4 - 9$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$y - fa = -2(x - a)$

$y - 3 = -2(x + 1)$

$= -2x - 2 + 3$

$y = -2x + 1$

double check!

$y + 5 = -2(x - 3)$

$-2x + 6 - 5$

$y = -2x + 1$

$y = -2x + 1$

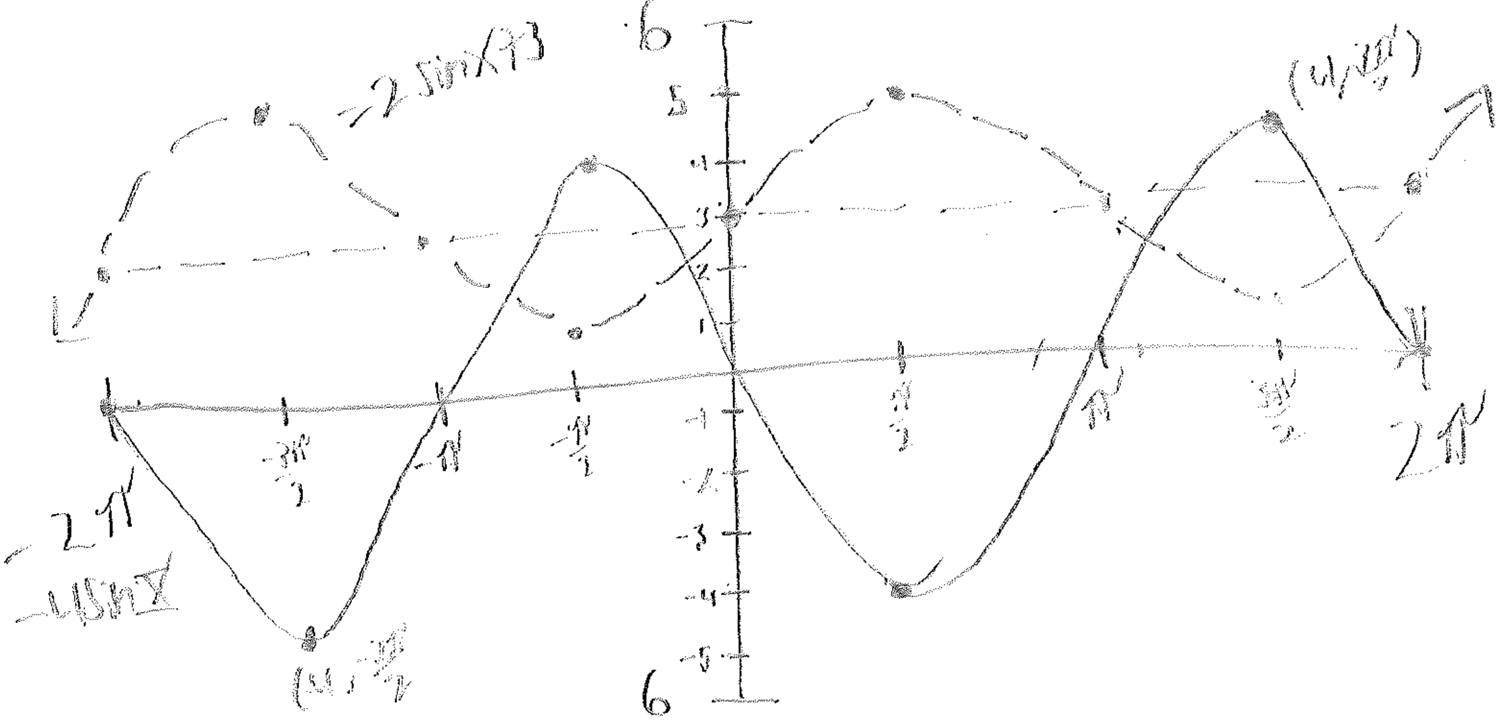
Sin =

-Sin =

2π ↑ $\quad 2\pi$ ↑

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$-4\sin = 2\sin + 3$

$-2\sin \quad -2\sin$

$-6\sin = 3$

$\overline{-6} \quad \overline{-6}$

$\sin = \frac{3}{-6} = -\frac{1}{2} =$ points at $\sin(-\frac{1}{2})$

$\boxed{\frac{7\pi}{6}, \frac{11\pi}{6}}$

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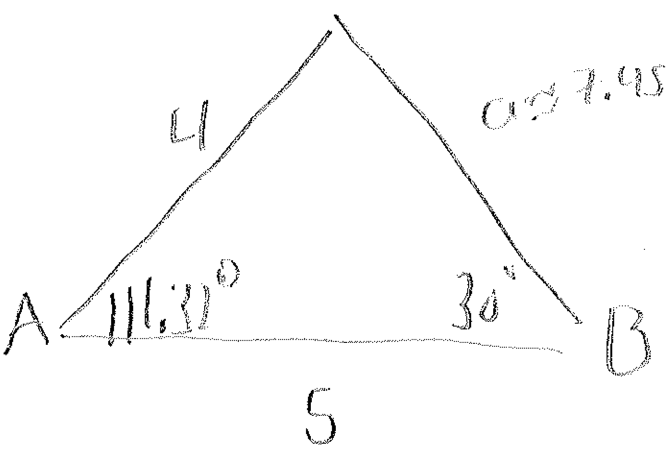
4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$180 - 30 - 38.68$

$b = 4, c = 5, B = 30°$

S S A

$C = 38.68$



$$\frac{\sin 30}{4} = \frac{\sin C}{5}$$

$$\sin^{-1}\left(\frac{5 \sin 30}{4}\right) = 38.68$$

1st Triangle
$C_1 \approx 38.68°$, $A_1 \approx 111.32°$
$a_1 \approx 7.45$

$a = \sqrt{5^2 + 4^2 - 2(5)(4) \cdot \cos(111.32)}$

$a = 7.45$

Solve 2nd Triangle

$180 - 38.68$

there is a 2nd!

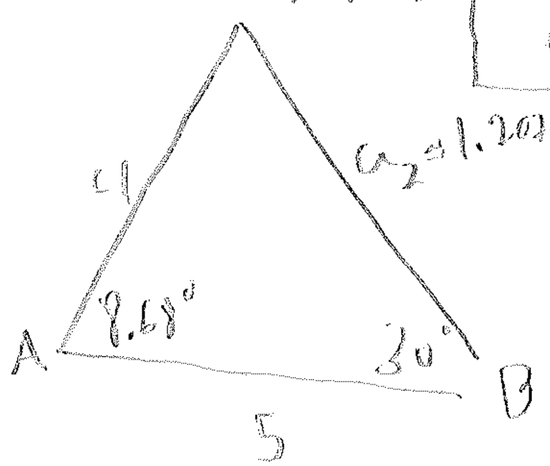
$141.32 + 30 = 171.32$

$180 - 171.32$

8.68

2nd Triangle
$C_2 \approx 141.32°$, $A_2 \approx 8.68°$
$a_2 \approx 1.207$

$C = 141.32°$

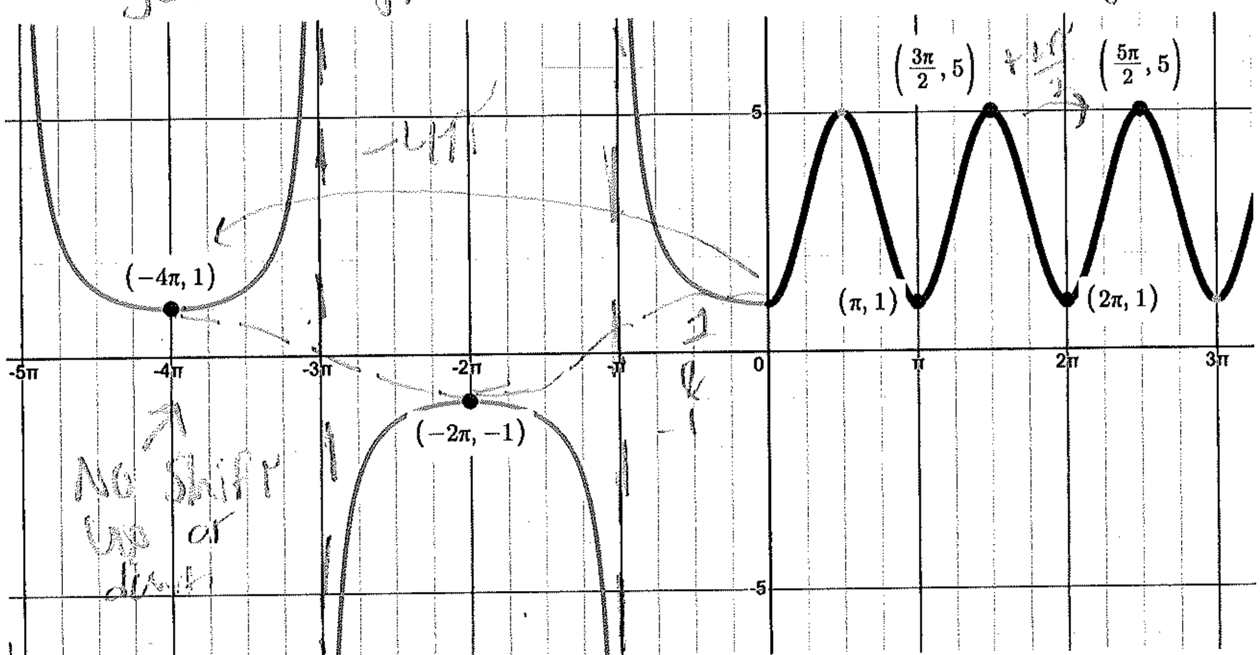


$a = \sqrt{5^2 + 4^2 - 2(5)(4)\cos(8.68)}$

$a = 1.207$

5) Given the graph below find the function $f(x)$ and state the domain and range.

Secant graph! -cosine graph



$\frac{2\pi}{-4\pi} = -\frac{1}{2}$

$\frac{2\pi}{1} \cdot \frac{1}{-4\pi}$

$\frac{2\pi}{\frac{2\pi}{2}}$

$\frac{2\pi}{1} \cdot \frac{2}{2\pi} = \frac{4}{2} = 2$

<0

$1\sec(-\frac{1}{2}x)$

range

$[-\infty, -1] \cup [1, \infty]$

Domain

$(-\infty, 0] \cap (K - \frac{2\pi}{1})$

>0

$2\cos(2x)+3$

range

$[1, 5]$

Domain

$[0, \infty)$

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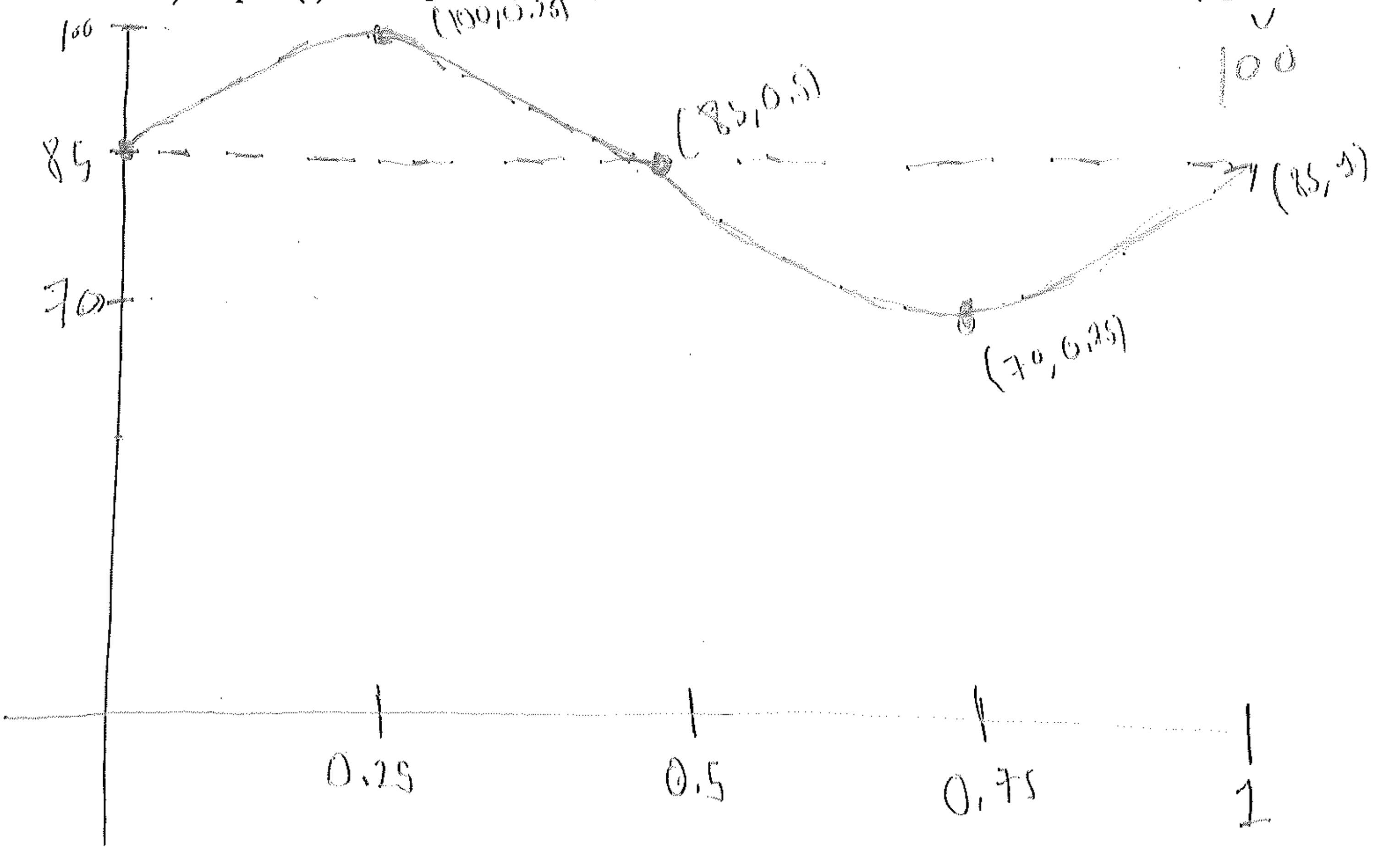
6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.

15+85
100



b) The systolic pressure is 100 BPM

c) The diastolic pressure is 70 BPM

d) The heart rate is 60 beats per minute

Midline = $\frac{100+70}{2} = 85$

systolic = 15+85
100

diastolic = 85-15
70

period = $\frac{2\pi}{B} = \frac{2\pi}{2\pi} = 1$

heart rate = $\frac{60}{\text{period}} = \frac{60}{1} = 60$

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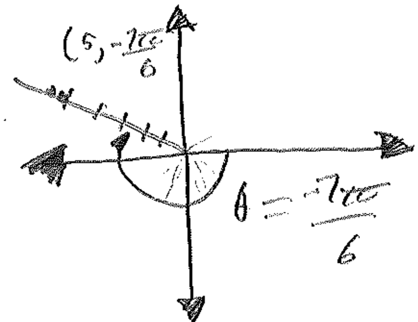
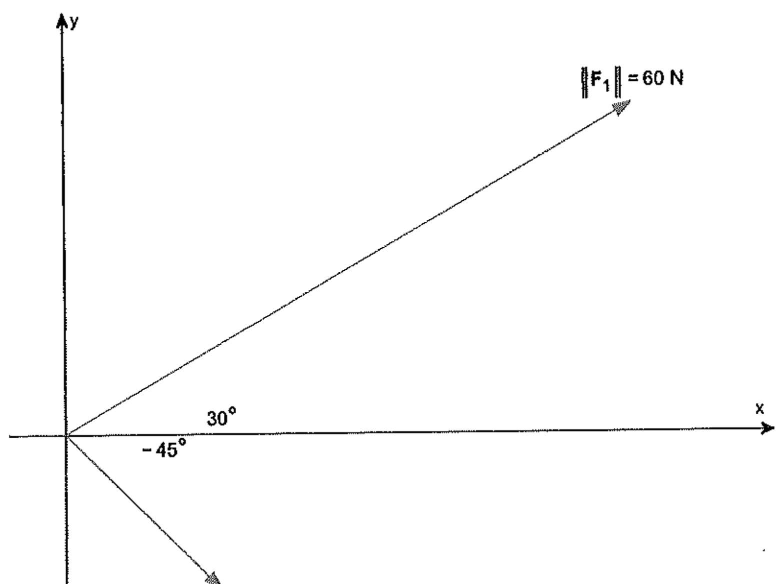
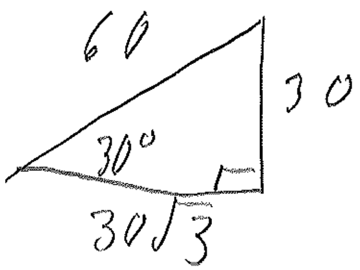
1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

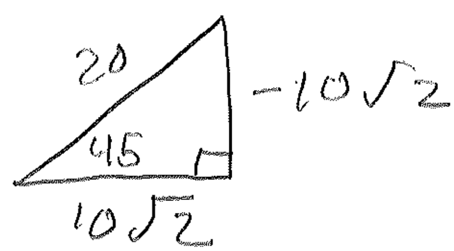
$\frac{7\pi}{6}$



1b) Add the two vectors $F_1 + F_2$







$F_1 + F_2 = 30\sqrt{3} + 10\sqrt{2}\, i + 30 - 10\sqrt{2}\, j$

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2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

$a \longrightarrow b$

$-1 \longrightarrow 3$

$$\frac{f(b)-f(a)}{b-a}$$

$\underline{a}$ $\quad f(-1) = 3$ (x_1, y_1)

$4-(-1)^2$

$4-1=3$

$\underline{b}$ $\quad f(3) = -5$ (x_2, y_2)

$4-(3)^2$

$4-9=-5$

$$\frac{-5-(3)}{3-(-1)}$$

$$\frac{-8}{4} = -2$$

$m=-2$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$y-y_1=m(x-x_1)$

$y-(3)=-2(x-(-1))$

$y-3=-2(x+1)$

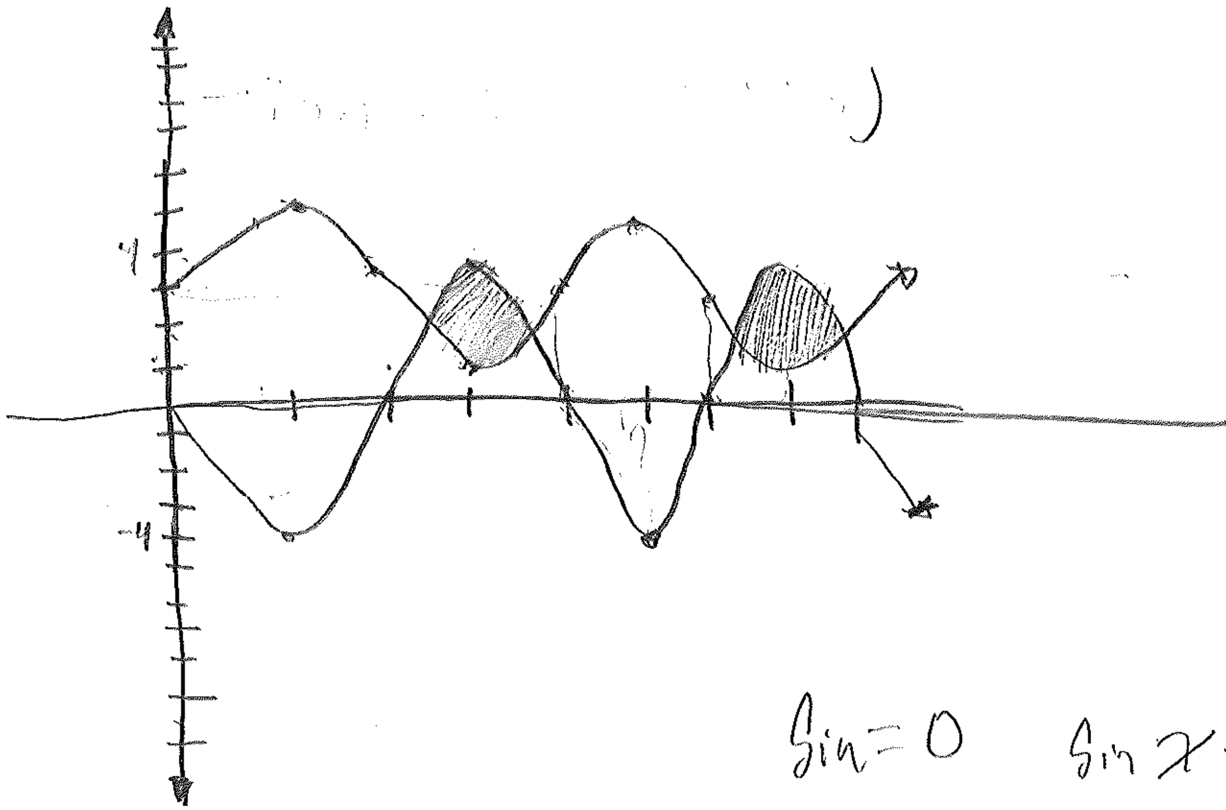
$y-3=-2x-2$

$+3 \qquad\qquad +3$

$y=-2x+1$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



$\sin = 0$ $\quad \sin x - \frac{3}{6}$

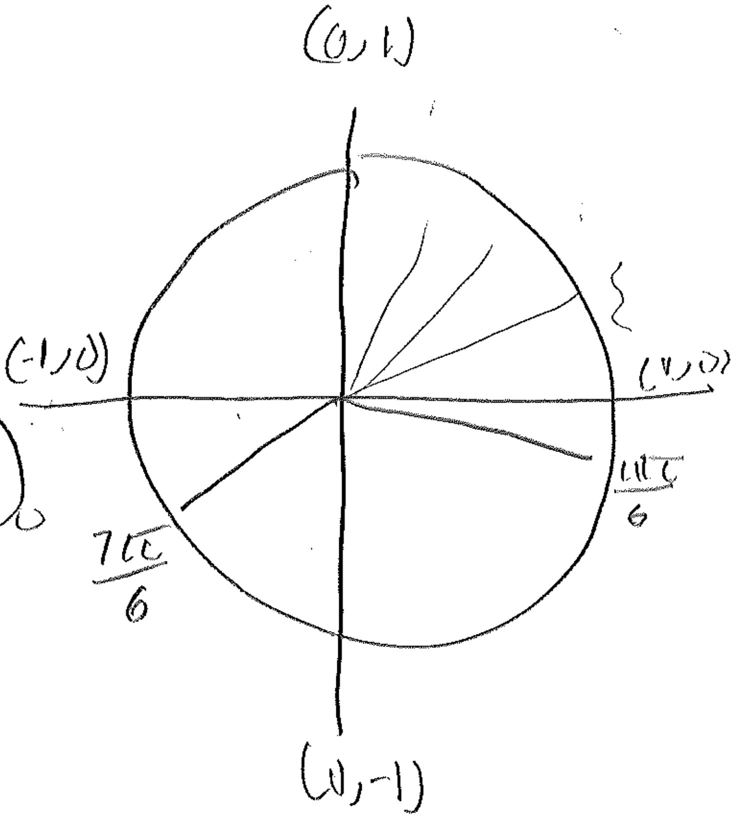
Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$-4\sin = 2\sin(\cancel{x}) + 3$
$-2 \quad -2\sin(\cancel{x})$

$\frac{-6\sin}{-6} = \frac{3}{-6} \quad \frac{}{-6}$

$\sin = \frac{3}{6} = -\frac{1}{2}$

$\left\{ \frac{7\pi}{6}, \frac{11\pi}{6} \right\}$



Name ________________________________ ID# ________________________________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30^\circ$$

$$\frac{\sin 30^\circ}{4} = \frac{\sin C^\circ}{5}$$

$$\frac{5 \sin 30^\circ}{4} = \frac{4 \sin C'}{4}$$

$$\sin C^{-1}\left(\frac{5 \sin 30^\circ}{4}\right) \approx 38.682$$

$$38.68 + 30 = 68.68$$

$$180 - 68.68$$

$C_1 \approx 38.682$

$C_2 \approx 141.318$

$A_1 \approx 111.32$

$A_2 \approx 8.69$

$a_1 \approx 1.05703$

$a_2 \approx 1.0208706$

A_1

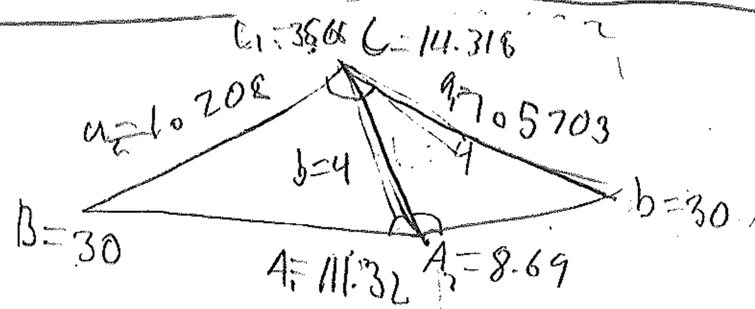
$$\frac{\sin 111.32}{a} = \frac{\sin 30}{4}$$

$$a_1 = \frac{4 \sin 111.22}{\sin 30}$$

A_2

$$\frac{\sin 8.69}{a} = \frac{\sin 30}{4}$$

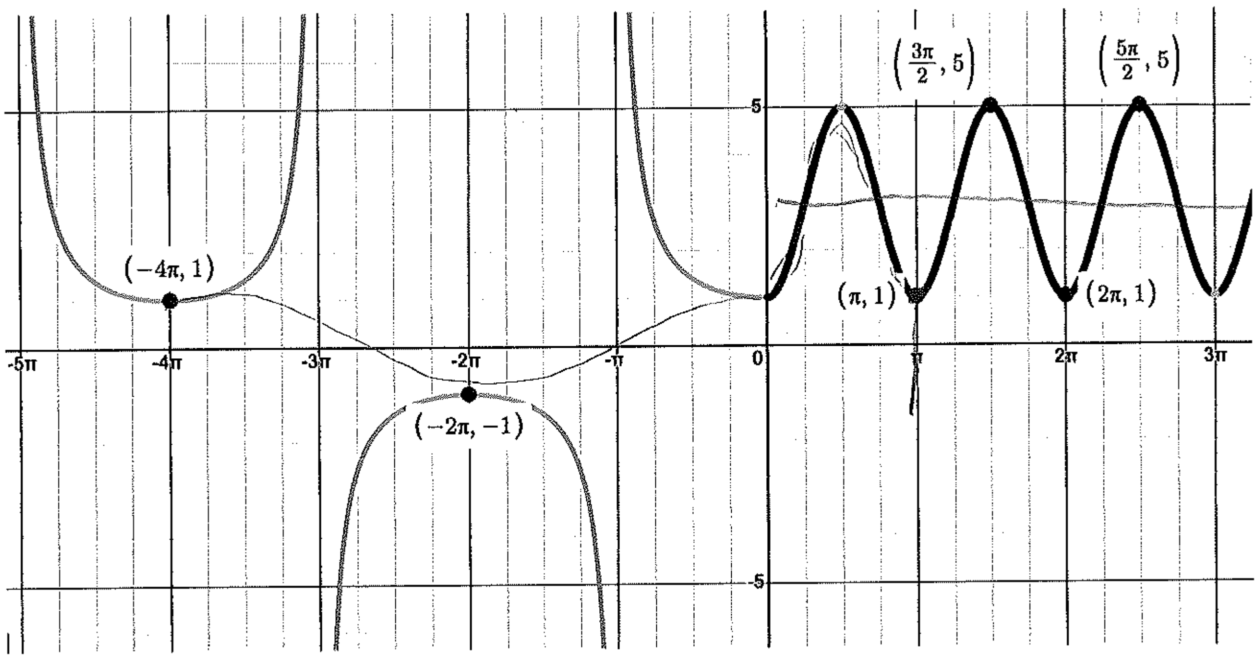
$$a_2 = \frac{4 \sin 8.69}{\sin 30^\circ}$$



$\frac{2\pi}{\frac{\pi}{2}}$ $\frac{2\pi}{1}\cdot\frac{2}{\pi}$ $\frac{4}{1}$ $\frac{2\pi}{1}\cdot\frac{1}{\pi}$ $= 2\pi$ $\frac{2}{1}$ cos

5) Given the graph below find the function $f(x)$ and state the domain and range.

$\sec\left(\frac{\pi}{4}x\right)$ $\quad -2\cos(\pi x)+3$



$\frac{-4}{\pi}$ $\quad \frac{\pi}{1}\cdot\frac{1}{4}$ $\quad \frac{\pi}{4}$

Domain $(-\infty, \infty)$

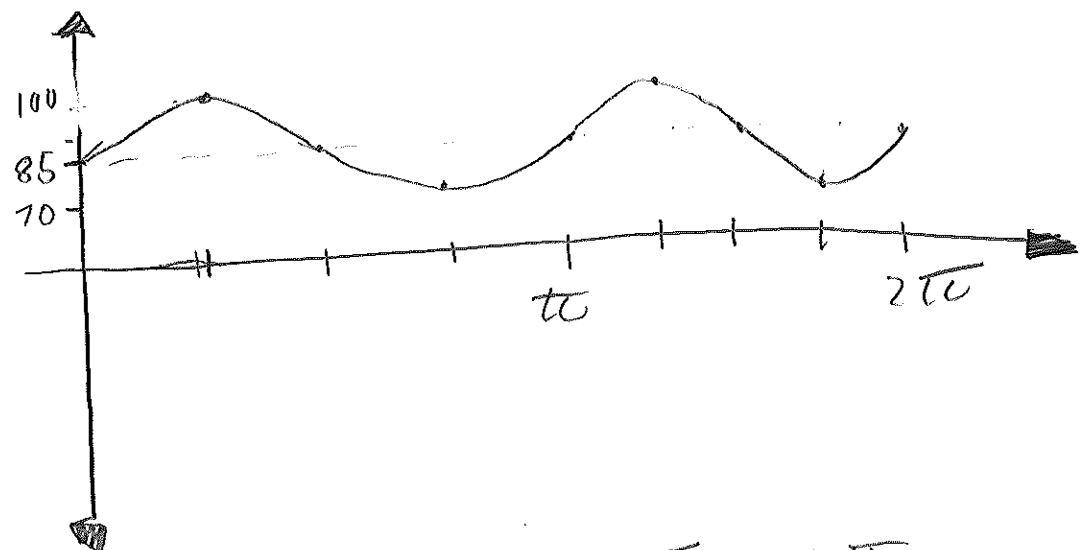
range $(-\infty, -1] \cup [1, \infty)$

$\frac{2\pi}{2}$ $\frac{2\pi}{1} \cdot \frac{1}{2\pi}$ $\frac{2}{2} = \pi$

Name ______________________________ ID# ______________________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



$\frac{\pi}{1} \cdot \frac{2\pi}{1} = \frac{2\pi}{1}$

60

$\frac{60}{1}$ $\frac{2\pi}{1} = 120$

b) The systolic pressure is 100

c) The diastolic pressure is 70

d) The heart rate is 120 beats per minute

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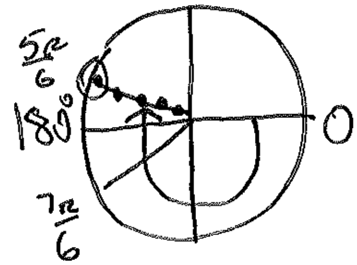
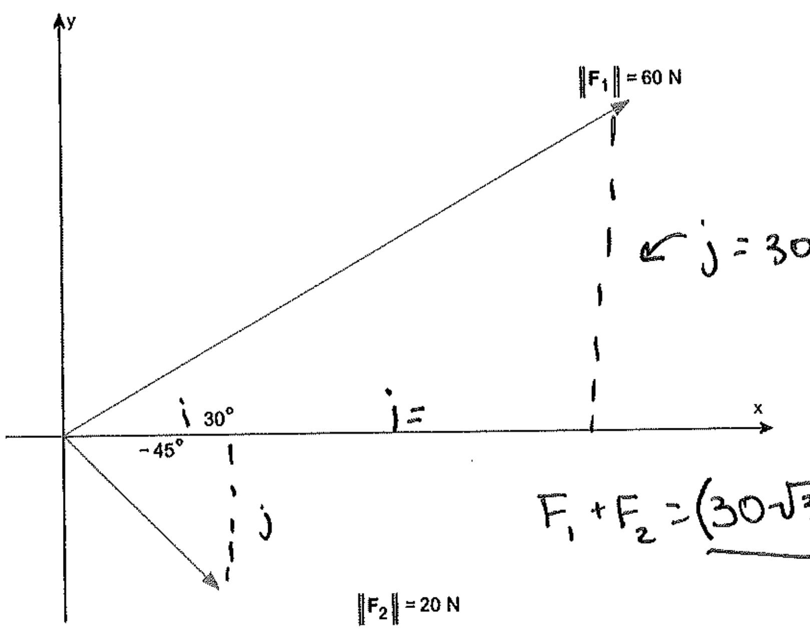


1b) Add the two vectors $F_1 + F_2$

Soh cah toa



$F_1 + F_2 = \boxed{(30\sqrt{3} + 10\sqrt{2})i + (-10\sqrt{2} + 30)j}$

$F_1)$ $\sin 30° = \frac{j}{60} \rightarrow \frac{1}{2} = \frac{j}{60} \rightarrow \frac{60}{2} = \frac{2j}{2} \rightarrow \boxed{j = 30}$

$\cos 30° = \frac{i}{60} \rightarrow \frac{\sqrt{3}}{2} = \frac{i}{60} \rightarrow 2i = 60\sqrt{3} \rightarrow \boxed{i = 30\sqrt{3}}$

$F_1 = 30\sqrt{3}i + 30j$

$F_2)$ $\sin -45° = \frac{j}{20} \rightarrow \frac{-\sqrt{2}}{2} = \frac{j}{20} \rightarrow \frac{2j}{2} = \frac{-20\sqrt{2}}{2} \rightarrow \boxed{j = -10\sqrt{2}}$

$\cos -45° = \frac{i}{20} \rightarrow \frac{\sqrt{2}}{2} = \frac{i}{20} \rightarrow \frac{2i}{2} = \frac{20\sqrt{2}}{2} \rightarrow \boxed{i = 10\sqrt{2}}$

$F_2 = 10\sqrt{2}i - 10\sqrt{2}j$

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2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

$$\frac{f(b) - f(a)}{b - a}$$

$f(b) = 4 - 3^2 = 4 - 9 = -5$

$f(a) = 4 - (-1)^2 = 4 - 1 = 3$

$$\frac{-5-3}{3-(-1)} = \frac{-2}{4} = -\frac{1}{2}$$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

slope

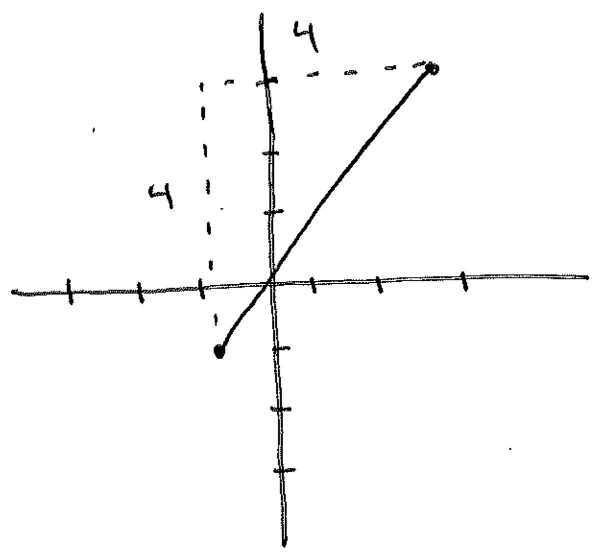
$\frac{4}{4} = 1$

$y - y_1 = m(x - x_1)$

$y - 3 = 1.(x - 3)$

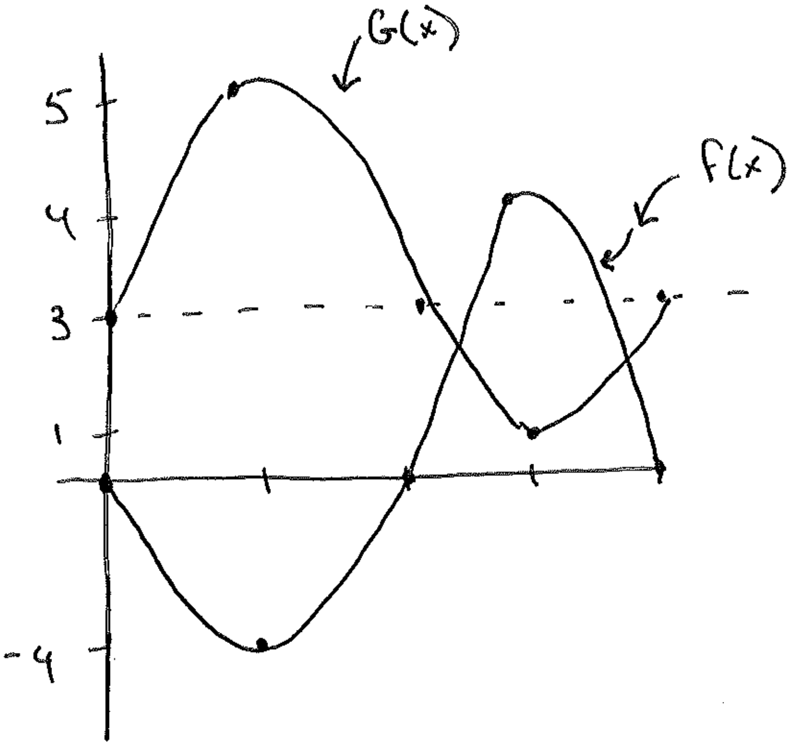
$+3 \qquad +3$

$y = 1x$



3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$-4\sin x = 2\sin x + 3$

$-2\sin$

$\dfrac{-6\sin x}{-6} = \dfrac{3}{-6}$

$\sin x = -\dfrac{3}{6}$

$\sin x = -\dfrac{1}{2}$

$\sin x = \dfrac{-1}{2}$ at $\dfrac{7\pi}{6}$ and $\dfrac{11\pi}{6}$

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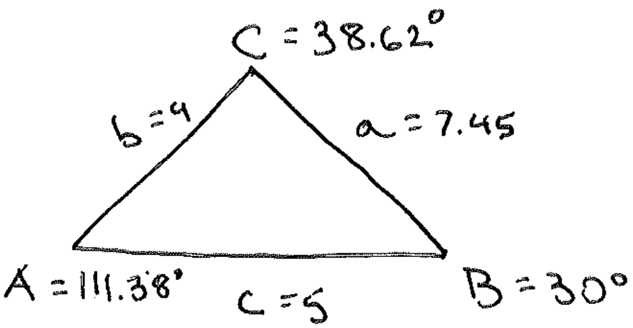
4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30°$$



b = 4
c = 5
B = 30°

SSA
↓
possibility of two triangles

Triangle 1	Triangle 2
$A° = 111.38$	$C° = 141.38$
$C° = 38.62$	$A° = 8.62$
$a = 7.45$	$a = 1.12$

$\frac{\sin 30°}{4} = \frac{\sin C}{5} \rightarrow \frac{5\sin 30°}{4} = \frac{4\sin C}{4}$

$\sin C = \frac{5\sin 30°}{4}$

$\sin C = \frac{2.5}{4}$ $\quad \sin C = 0.625$

$C° = \sin^{-1}(0.625)$

$\boxed{C° = 38.68}$

$180 - 38.62 - 30 = A°$

$A° = \boxed{111.38}$

$\frac{\sin 30°}{4} = \frac{\sin 111.38}{a}$

$a \sin 30° = 4 \sin 111.38$

$a = \frac{4\sin 111.38}{\sin 30°} \rightarrow \frac{1}{2}$

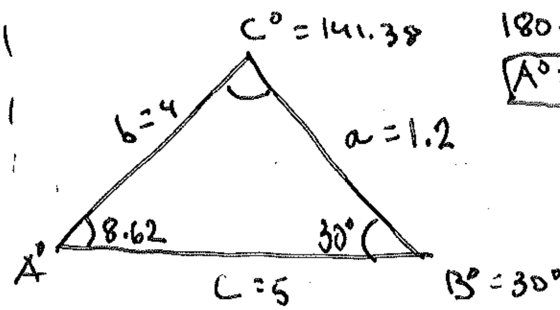
$a = \frac{3.72473}{\frac{1}{2}} \rightarrow \boxed{a = 7.4494}$

Triangle 2

$180° - 38.62 = \boxed{141.38 = C°}$

$180 - 141.38 - 30 = A°$

$\boxed{A° = 8.62}$

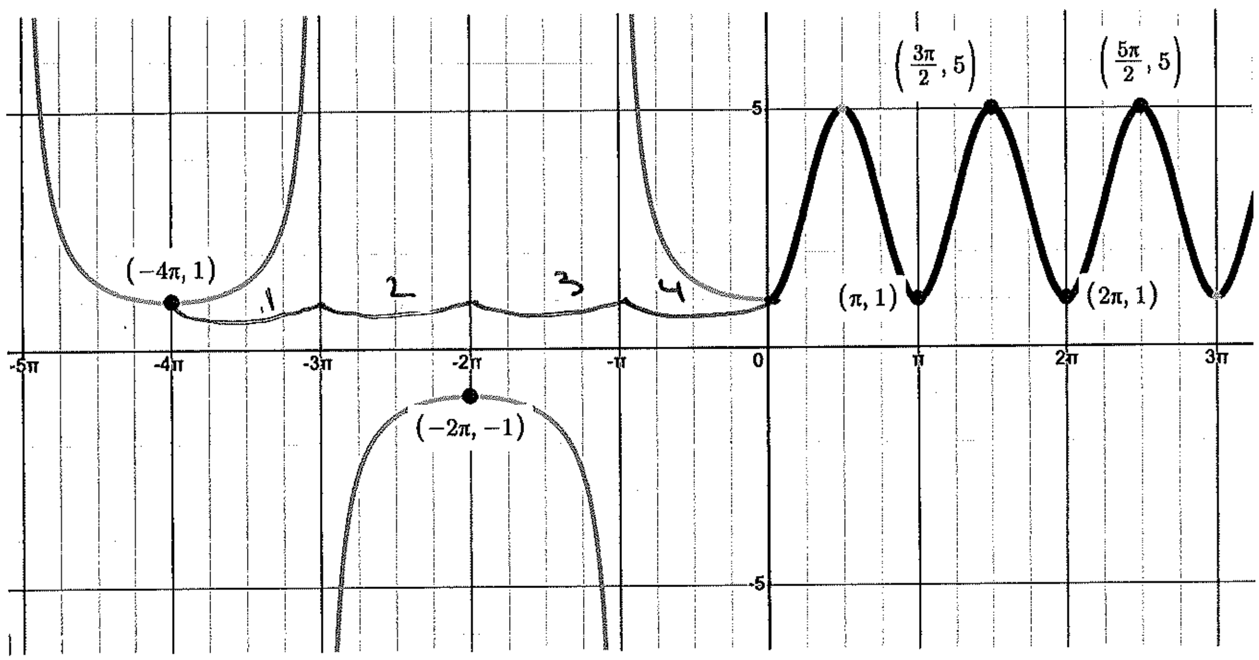


$a = \frac{4\sin 8.62}{\sin 30°} \rightarrow \frac{0.59952}{\frac{1}{2}}$

$\boxed{a = 1.19904}$

$\frac{\sin 30°}{4} = \frac{\sin 8.62}{a} \rightarrow \sin 30° a = 4\sin 8.62$

5) Given the graph below find the function $f(x)$ and state the domain and range.



for red graph

domain $= (\mathbb{R}, \neq 2\pi k+1) \cup [0, \infty)$

range $= (-\infty, -1] \cup [1, \infty)$

Period: $\frac{2\pi}{B} = \frac{4\pi}{1} \rightarrow \frac{2\pi}{4\pi} = \frac{4\pi B}{4\pi} \rightarrow B = \frac{1}{2}$

So, it's $\sec\left(\frac{1}{2}x\right)$

for black graph $\longrightarrow$ So, $2\sin(2x)+3$

amplitude $= \frac{5-1}{2} = \frac{4}{2} = 2$

midline $= \frac{5+1}{2} = \frac{6}{2} = 3$ (vertical shift)

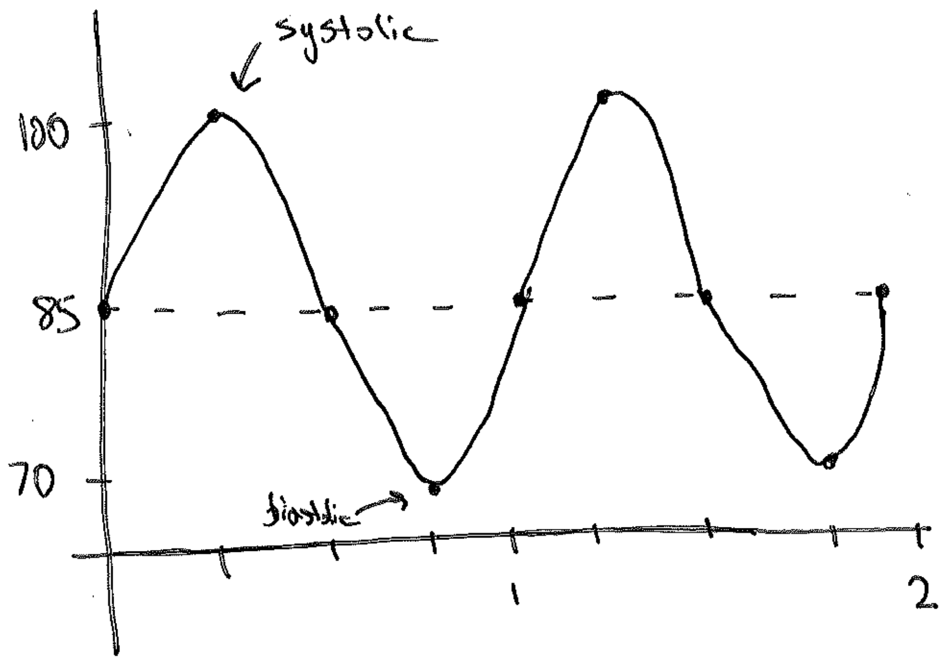
Period is $\frac{5\pi}{2} - \frac{3\pi}{2} = \frac{2\pi}{2} = \pi \rightarrow \frac{2\pi}{B} = \frac{\pi}{1} = \frac{\pi B}{\pi} = \frac{2\pi}{\pi} = 2$

Name ID#

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.

$\frac{2\pi}{2\pi} = 1$ ← period



b) The systolic pressure is 100

c) The diastolic pressure is 70

d) The heart rate is 60 beats per minute

heart rate is $\frac{1}{T} \rightarrow \frac{1}{1}$

$\frac{1}{1} \times \frac{60 \text{ seconds}}{1 \text{ minute}} = \frac{60}{1} = 60 \text{ bpm}$

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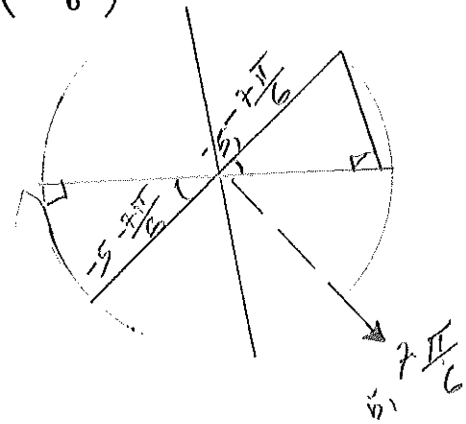
1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

R = 5 for the distance

for $\theta = -\frac{7\pi}{6}$ $\quad -1 \frac{7\pi}{6} = 210°$



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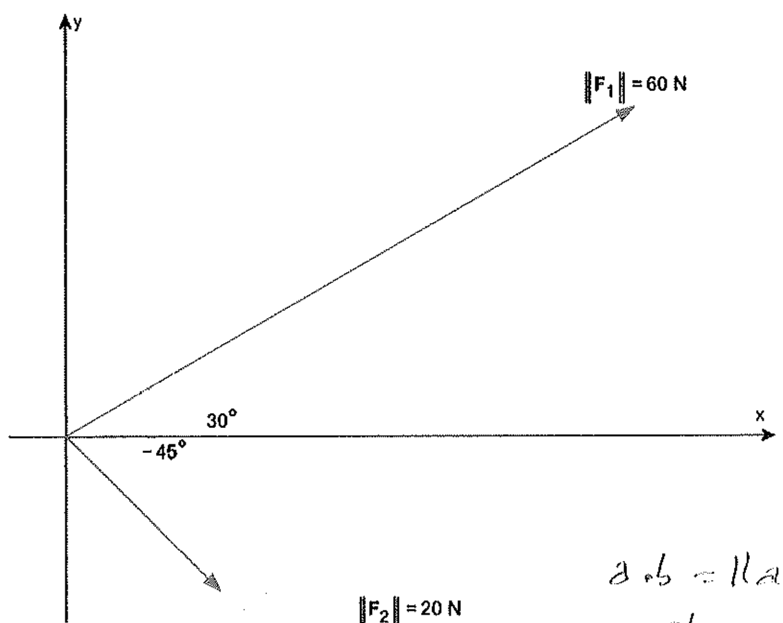


CROSS
V×W
V = 60 i - j

$a \cdot b = \|a\| \cdot \|b\| \cdot \cos\theta$

$\frac{ab}{\|a\| \cdot \|b\|}$

$a \cdot b = 0 \quad \|a\| \cdot \|b\|$

$60\,i + 20\,j\ ;\ 20\,i - 60\,i$

$\langle 60, 20\rangle \cdot \langle 20 - 60)$

$60 \cdot 20 + 20(-60)$

$1200 - 1200 = 0$

Name ______________________ ID# ______________________

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

$g(x) = 4 - x^2$ from $x = -1$ to $x = 3$ thats the one I wanna use

$$\frac{g(3) - g(-1)}{3 - (-1)}$$

$$\frac{-5 - 3}{3 - (-1)} = \frac{-8}{4} = -2$$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$(-1, g(-1))$

$$m = \frac{-5 - 3}{3 - (-1)} = \frac{-8}{4} = -2$$

$y = -2x + 1$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.

$f(x) = -4 \sin(x)$

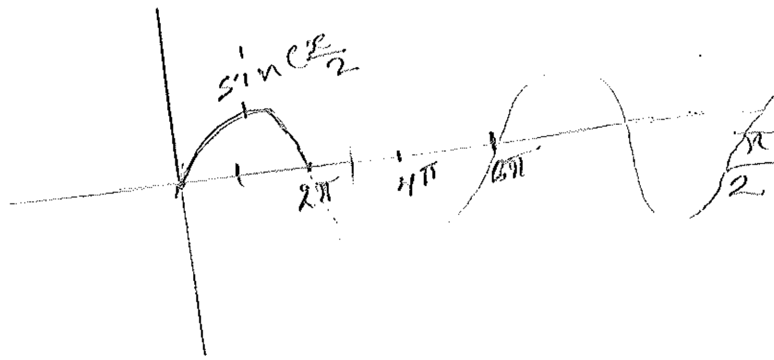
$g(x) = 2x \sin(x)+3$

$f(x)$ max $=0$, min. $=-4$

$g(x)$ max $=2+3=5$ min.

$=-2+3=1$

$T = \frac{2\pi}{1} = 2\pi$



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$f(x) = g(x)$ on the interval $[0, 2\pi]$

$f(x) = -4 \sin(x)$

$g(x) = 2\sin(x)+3$

$\sin(x) = -\frac{1}{2}$ in $[0, 2\pi]$

$x = \frac{7\pi}{6}$ $\quad x = \frac{11\pi}{6}$

Name ______________________________ **ID#** ______________________________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30°$$

$\frac{\sin A}{a} = \frac{\sin B}{b}$

$a^2 = b^2 + c^2 - 2bc \cdot \cos A$

$8 = 4^2 + 5^2 - 8 \cdot 4 \cdot 5$

$16 = 16 + 25 - 160 \cos A$

$16 = 41 - 160 \cos A$

$16 = 160 \cos A - 4.1$

-41

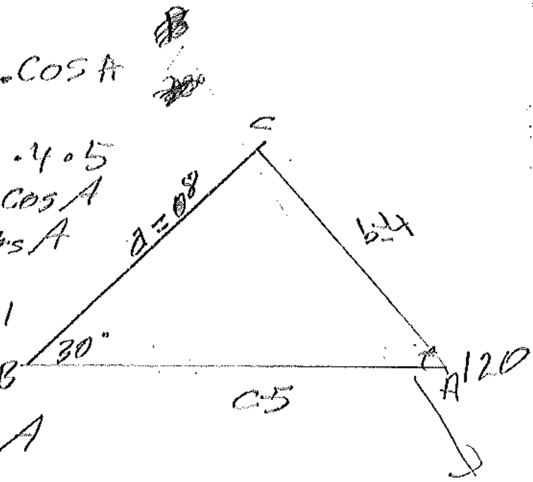
$-25 = -160 \cdot \cos A$

$\frac{-25}{160} \cos A$

$\cos A$

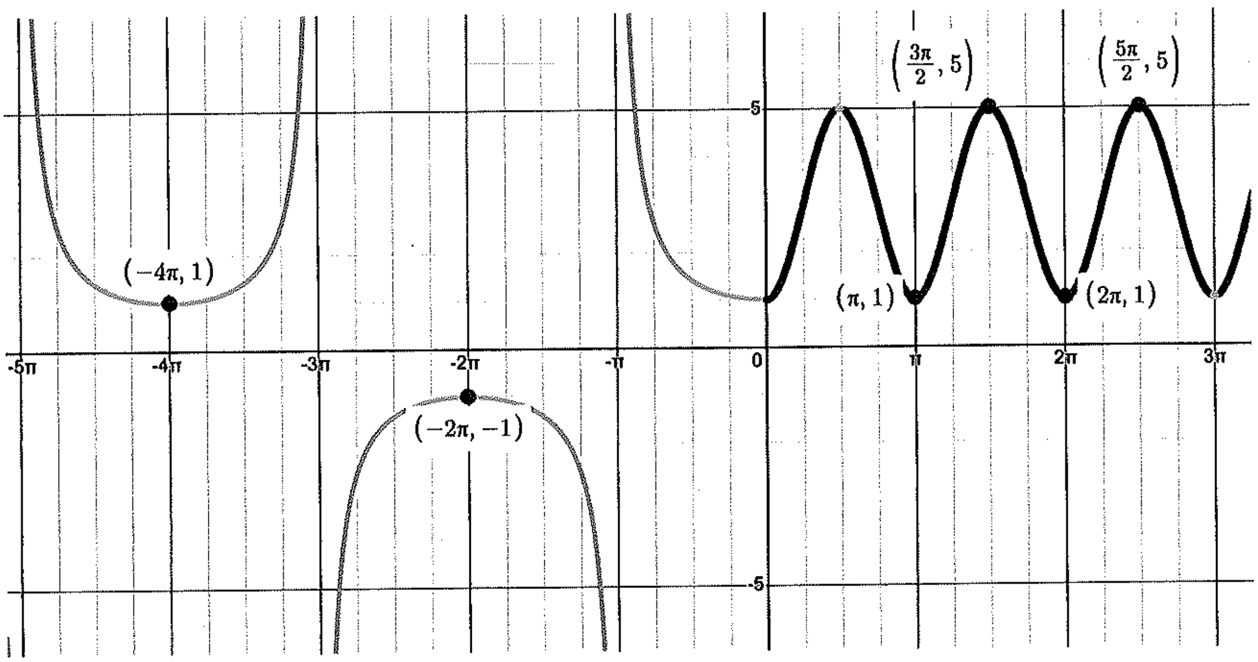
$\frac{25}{80} = \cos A$

$\cos^{-1}\left(\frac{37}{160}\right)$



$\cos^{-1}\left(\frac{37}{80}\right)$

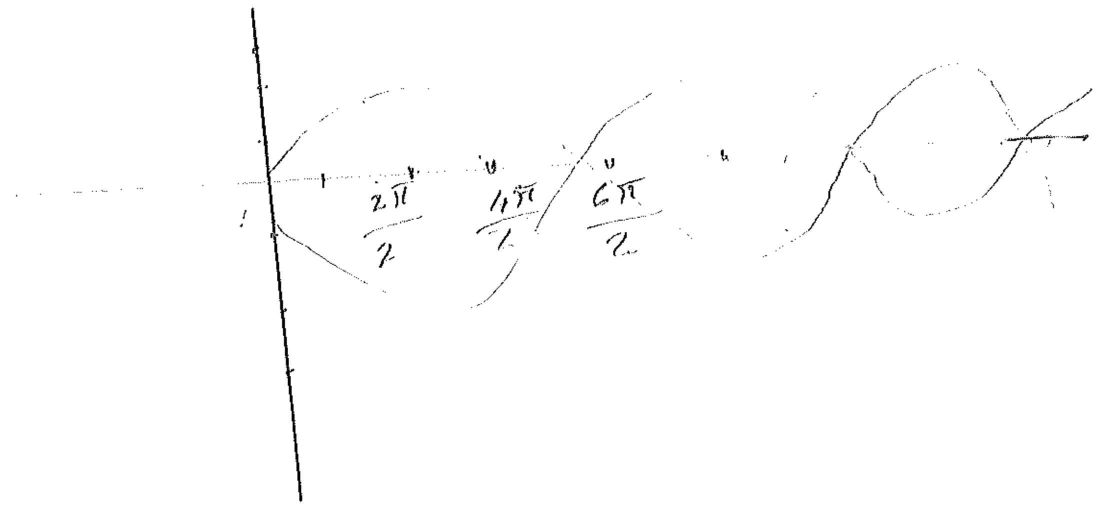
5) Given the graph below find the function $f(x)$ and state the domain and range.



Name ______________________ ID# ______________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



b) The systolic pressure is 75

c) The diastolic pressure is 85

d) The heart rate is 15 beats per minute

Name [redacted] **ID#** [redacted]

Math 104 Spring 2025 Final Exam

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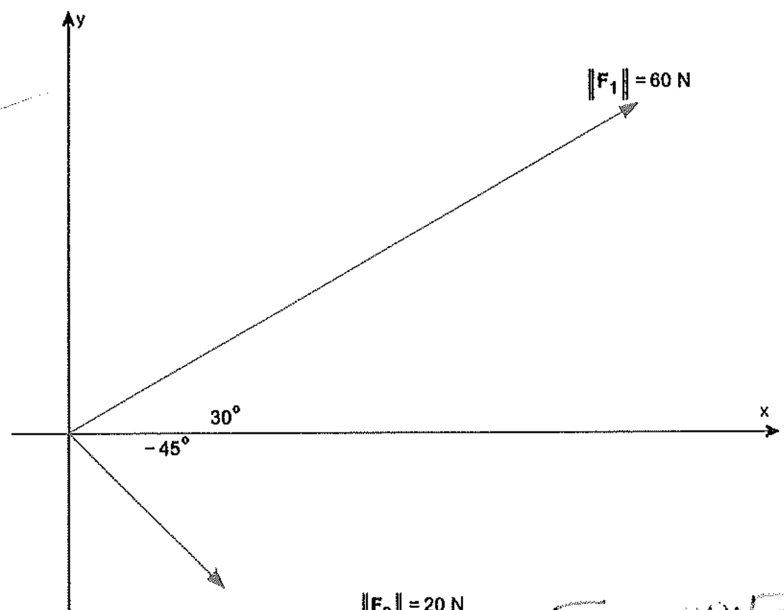
Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

$x = r\cos\theta$
$x = 5\cos(\frac{-7\pi}{6})$.
$x = 5(-\frac{\sqrt{3}}{2}) =$

$y = r\sin\theta$
$y = 5\sin(\frac{-7\pi}{6})$
$y = 5(-\frac{1}{2}) . =$

1b) Add the two vectors $F_1 + F_2$



$x = |F_1|\cos(\theta) = 60 \cdot \cos(30°) = 60 \cdot \frac{\sqrt{3}}{2} = 30\sqrt{2}\ N$
$x = |F|\cos(\theta) = 20 \cdot \cos(-45°) = 20 \cdot \frac{\sqrt{2}}{2} = 10\sqrt{2}\ N.$
$y = |F_1|\sin(\theta) = 60 \cdot \sin(30°) = 60 \cdot \frac{1}{2} = 30N$
$y = |F_2|\sin(\theta) = 20 \cdot \sin(-45°) = 20 \cdot -\frac{\sqrt{2}}{2} = -10\sqrt{2}$

$F_1 + F_2 = (30\sqrt{2}\ N + 10\sqrt{2}\ N)i + (30N - 10\sqrt{2}\ N)j$

Name ______________________ ID# ______________________

$\quad\quad\quad\quad\quad\ \ a \quad b$

2) Let $g(x) = 4 - x^2$.

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad a \quad b$

a) Find the average rate of change from -1 to 3.

$\frac{f(b)-f(a)}{b-a}$ $\quad \frac{f(3)-f(-1)}{3-(-1)}$

$f(3) = 4-(3)^2 = -5$ $\quad \frac{-5-3}{3-(-1)} = \frac{-8}{4} = -2$

$f(-1) = 4-(-1)^2 = 3$

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad x_1 \quad y_1 \quad\quad x_2 \quad y_2$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

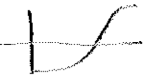
$y - y_1 = m(x - x_1)$ $\quad y-(-1) = -2(x-(-1))$

$m = -2$ $\quad y-(-1) = -2(x+1)$

$y-(-1) = -2x-2$

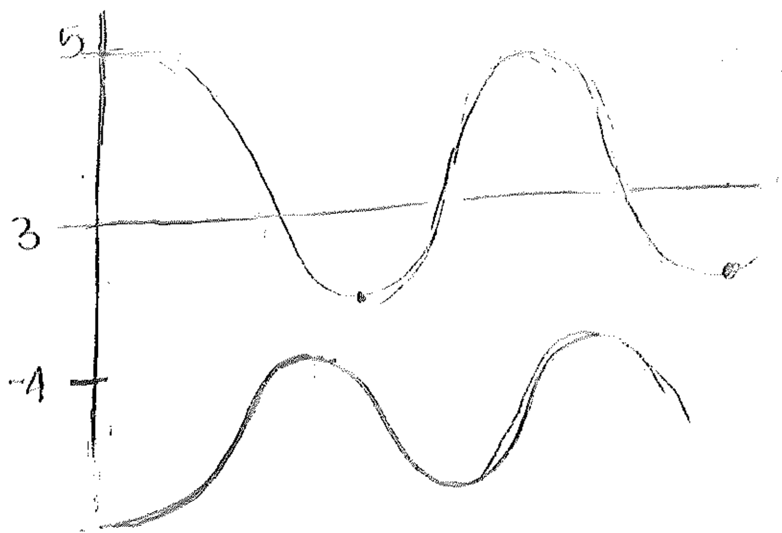
$y = -2x-2-1$

$y = -2x-3$

$\sin(x)$:  $-\sin(x)$ 

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



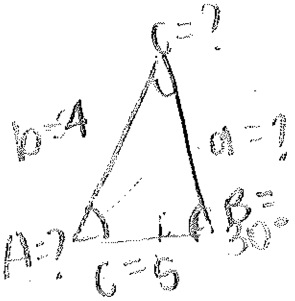
Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$-4\sin(x) = 2\sin(x) + 3$

Name ______________________ ID# ______________________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$b = 4, c = 5, B = 30°$ SSA



$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$\frac{\sin(30°)}{4} = \frac{\sin C}{5} = \frac{5\sin(30°)}{4} = 0.625$

$\sin^{-1}(0.625) = \underline{38.68}$

$38.68 + 30 + \underline{111.32} = 180°$

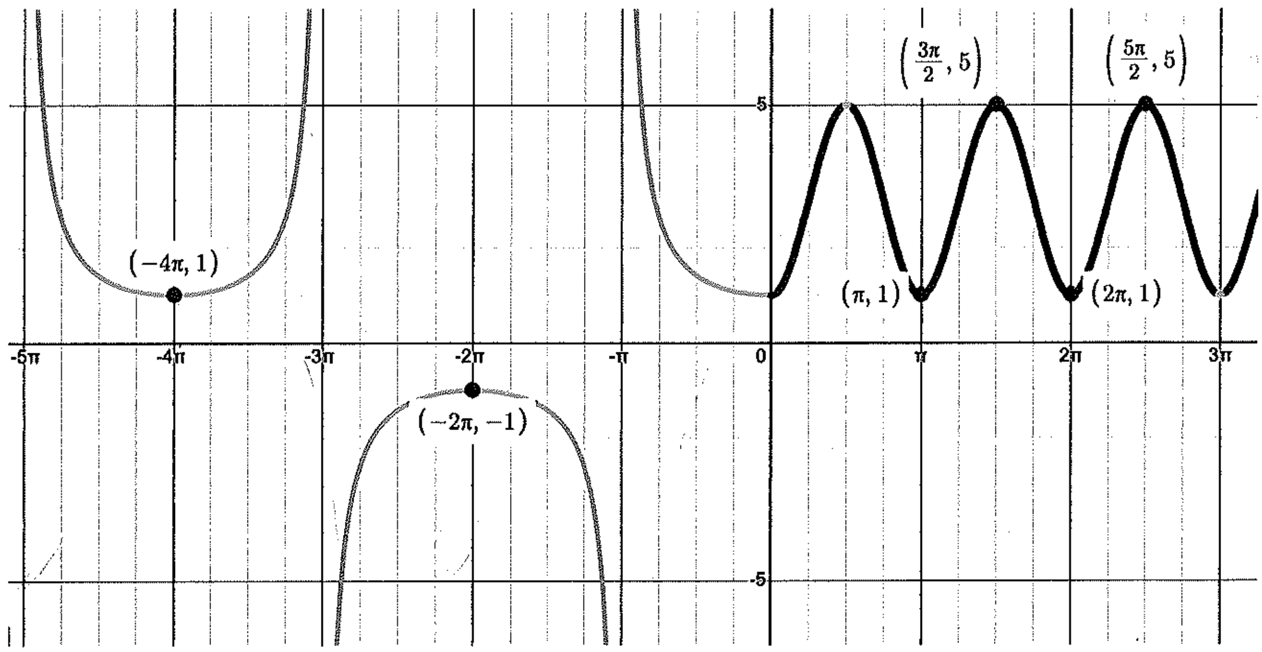
$a = \frac{4\sin(111.32°)}{\sin(30°)} = \frac{3.726}{0.5} = 7.45$

A = 111.32
C = 38.68
c = 7.45

Results in one triangle.

5) Given the graph below find the function $f(x)$ and state the domain and range.

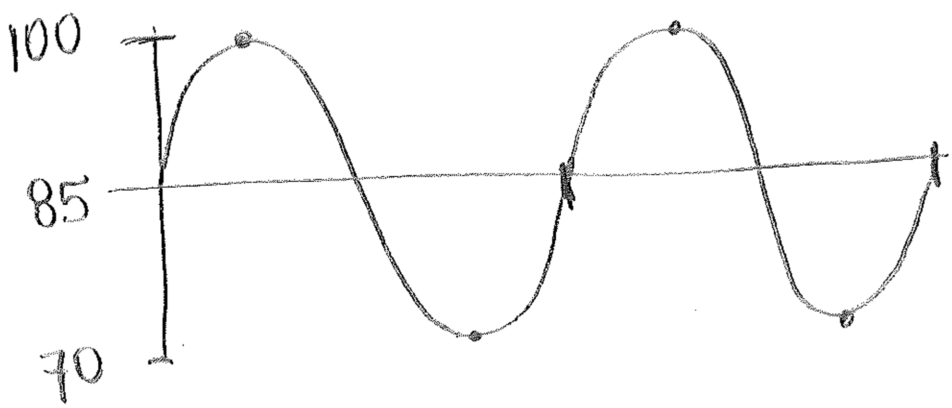
$f(x) = -4\sec$
$g(x) = 5\cos(\pi)$



Name ______________________ **ID#** ______________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



$$\frac{2\pi}{\frac{2\pi}{3}} = \frac{2\pi}{1} \cdot \frac{3}{2} = \frac{6}{2}$$

$$\frac{60}{\frac{6}{2}} = \frac{60 \cdot 2}{6} = 20 \text{ BPM}$$

$$HR = \frac{60 \text{ sec}}{1 \text{ min}} * \frac{1 \text{ beat}}{\frac{6}{2} \text{ sec}}$$

b) The systolic pressure is 100

c) The diastolic pressure is 70

d) The heart rate is 20 beats per minute

Name [redacted] **ID#** [redacted]

Math 104 Spring 2025 Final Exam

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Question		Points
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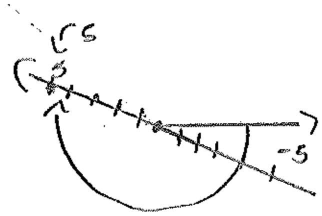
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

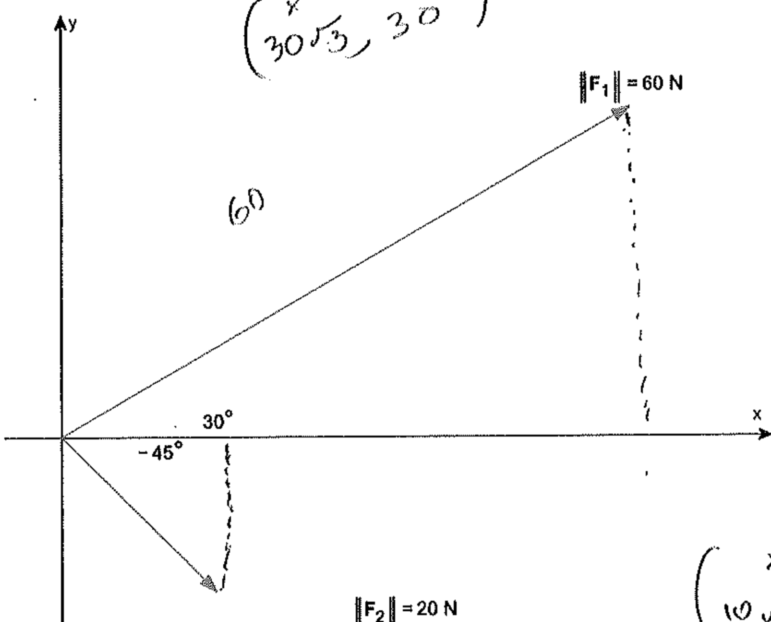
1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$

$30° = \pi/6 = \sqrt{3}/2, 1/2 \to 60 \cdot \frac{\sqrt{3}}{2} = 30\sqrt{3}$ (x), $60 \cdot \frac{1}{2} = 30$ (y)

$(30\sqrt{3}, 30)$



$(10\sqrt{2}, -10\sqrt{2})$

$-45° = \frac{7\pi}{4}$

$\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}$

(x) $20 \cdot \frac{\sqrt{2}}{2} = 10\sqrt{2}$ $\quad 20 \cdot -\frac{\sqrt{2}}{2}$ $\quad -10\sqrt{2}$

$$\boxed{F_1 + F_2 = (30\sqrt{3} + 10\sqrt{2})i + \left(30 - \frac{\sqrt{2}}{2}\right)j}$$

Name ______________________ ID# ______________________

2) Let $g(x) = 4 - x^2$.

A B

a) Find the average rate of change from -1 to 3.

$$\frac{g(b) - g(a)}{b - a}$$

$B = 4 - (3)^2 = -5$

$A \quad 4 - (-1)^2 = 3$

$3, (-5)$

$$\frac{-5 - 3}{3 + 1}$$

$$\frac{-8}{4} = \boxed{-2}$$

Avg Rate of change

(slope)

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

Point Slope Formula

$y - y_1 = m(x - x_1)$

$(-1, 3)$

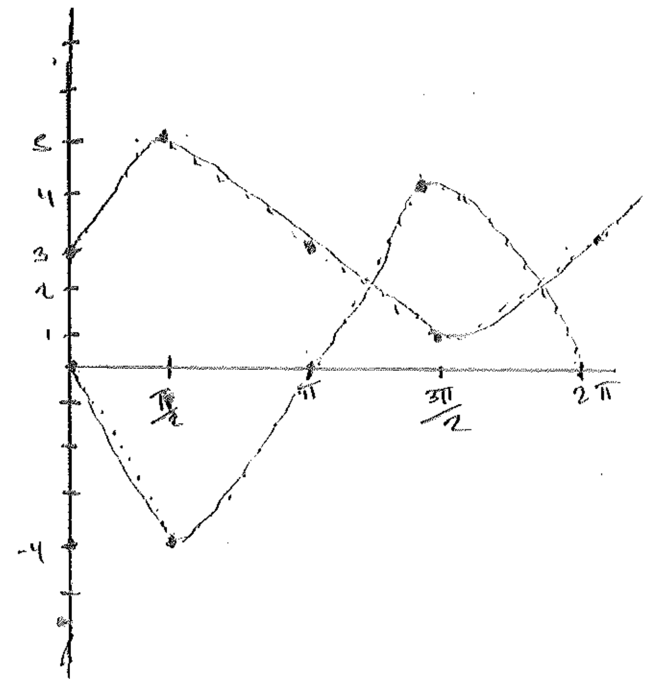
$y + 3 = -2(x + 1)$

$y + 3 = -2x - 2$
$\quad -3 \qquad\quad -3$

$$\boxed{y = -2x - 5}$$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$-4\sin(x) = 2\sin(x) + 3$

$+4\sin x \qquad +4\sin x$

$0 = 6\sin x + 3$

$-3 \qquad\qquad -3$

$\frac{-3}{6} = \frac{6\sin x}{6}$

$-\frac{1}{2}$

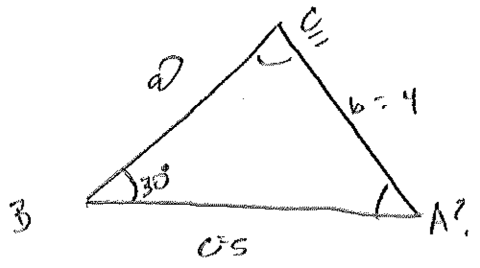
$\sin(x) = -\frac{1}{2}$

$\rightarrow \boxed{\frac{7\pi}{6} \ \& \ \frac{11\pi}{6}}$

Name ______________________ ID# ______________________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30^\circ$$



$C = 38.7^\circ$
$A = 111.3^\circ$
$a = 1.6$

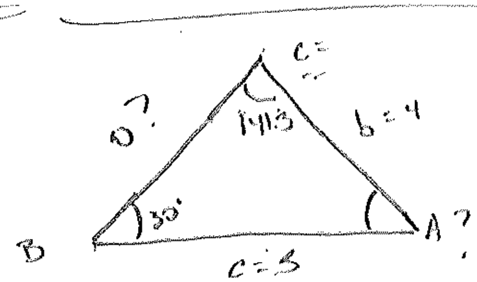
$$\frac{\sin 30^\circ}{4} = \frac{\sin C}{5} = \frac{5 \cdot \sin 30^\circ}{4} = \sin C \to \sin^{-1}\left(\frac{5 \cdot \sin 30^\circ}{4}\right) \cdot \frac{180}{\pi} = \underline{38.7^\circ}$$

$$\frac{\sin 30^\circ}{4} = \frac{\sin 11.3^\circ}{a} = a = \frac{4 \cdot \sin 11.3^\circ}{\sin 30^\circ} = \boxed{1.6} \quad \frac{\sin 38.7^\circ}{5} = \frac{\sin 11.3^\circ}{a} = 1.6\checkmark$$

2nd triangle check

$180 - 38.7 = 141.3$

$\to 180 - 141.3 - 30 = \underline{8.7^\circ}$ 2nd Triangle Available

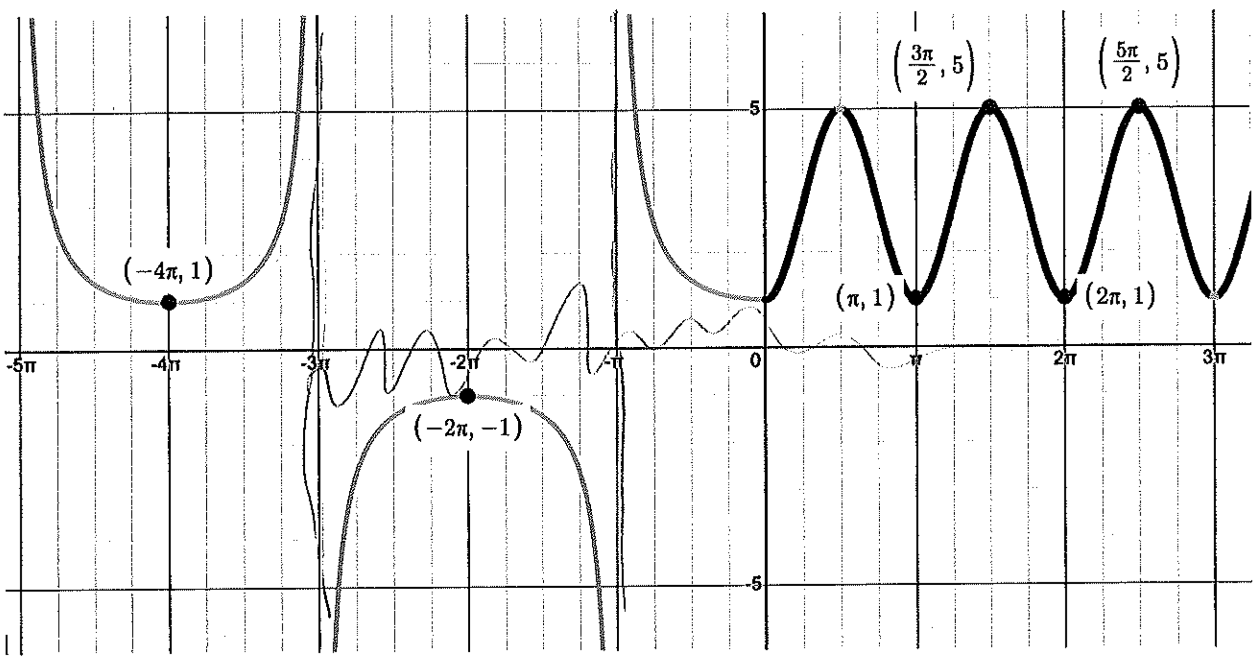


$C = 141.3$
$A = 8.7^\circ$
$a = 1.2$

$$\frac{\sin 30^\circ}{4} = \frac{\sin 8.7^\circ}{a} = \frac{4 \cdot \sin 8.7^\circ}{\sin 30^\circ} = 1.2$$

$$\frac{\sin 141.3}{5} = \frac{\sin 8.7^\circ}{a} = 1.2\checkmark$$

5) Given the graph below find the function $f(x)$ and state the domain and range.



$-2\csc\left(-\frac{1}{3}x\right)$

$\sin(2x)+1$

$\frac{\pi}{B}$ $\frac{\pi}{1}$ 3π

$\frac{2\pi}{1}$

$\frac{2\pi}{B} = \pi$

$\frac{2\pi}{\pi}$

$B = 2$

$f(x) = -2\csc\left(\frac{1}{3}x\right)$ when $(-\infty < x \leq 0)$

$F(x) = \sin(2x)+1$ when $(0 \leq x < \infty)$

Range $(-\infty, -1) \cup (1, \infty)$

Domain: all real #'s except $3\pi + k$ $\cup (0 \leq \infty)$

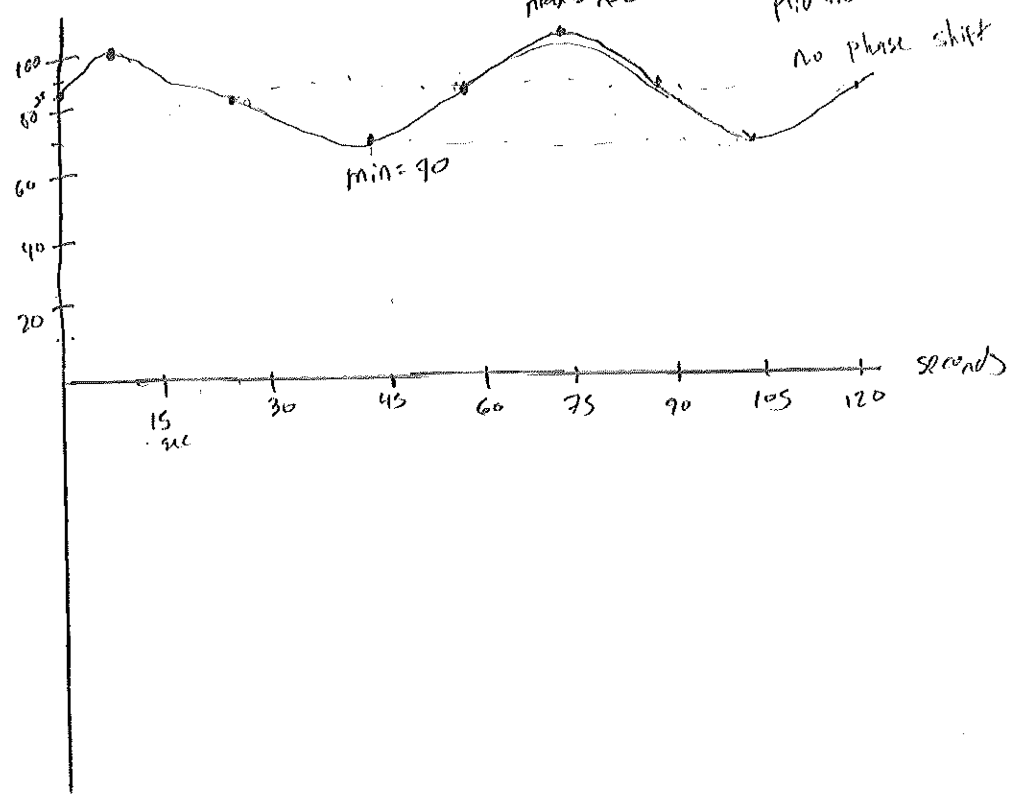
Name ______________________ ID# ______________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

$\frac{2\pi}{2\pi} = 1$ minute sec?

a) Graph $P(t)$ for two periods, label your axes and local maximums.

Amplitude = 15
Mid line = 85
no phase shift



b) The systolic pressure is 70

c) The diastolic pressure is 100

d) The heart rate is 30 beats per minute

Name [redacted] **ID#** [redacted]

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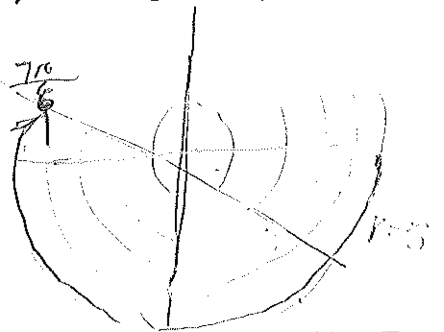
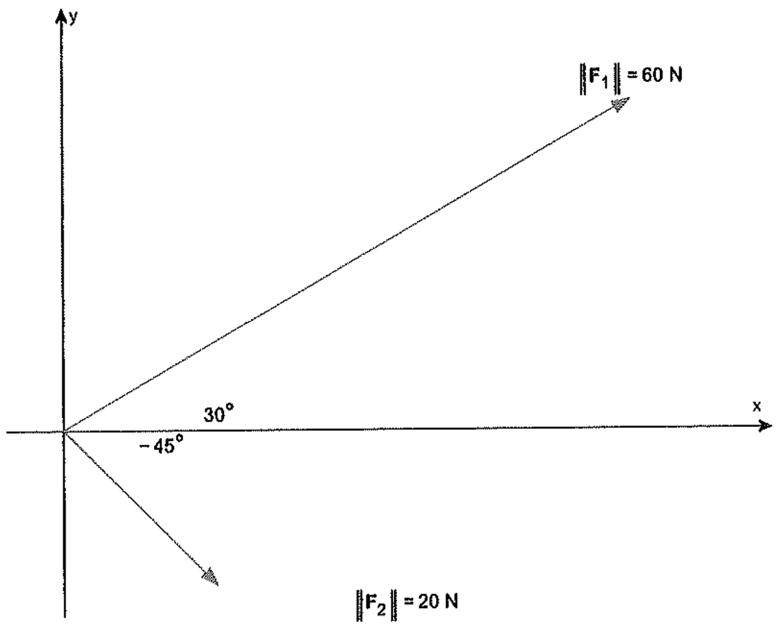
6	Applications	/6
Total		22/30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \dfrac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$



$F_1 = 60\angle 30°$
$F_1 = 60\cos(30°) = 60 \cdot \frac{\sqrt{3}}{2}$ $(30\sqrt{2})$
$F_1 = 60\sin(30°) = 60 \cdot \frac{1}{2} = 30$
$F_2 = 20_N - 45°$
$F_2 = 20\cos(-45°) = 20 \cdot \frac{\sqrt{2}}{2} = 10\sqrt{2}$
$F_2 = 20\sin(-45°) = 20(\frac{\sqrt{2}}{2}) = 10\sqrt{2}$

$F_1 + F_2 = (30\sqrt{3} + 10\sqrt{2})i +$
$((30 - 10\sqrt{2})j)$

Name _______________ **ID#** _______________

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

$m = \frac{f(b)-f(a)}{b-a} = \frac{-5-3}{3-(-1)} = \frac{-8}{4} = -2$

$b)$: $g(3) = 4-(3)^2$

$g(3) = 4-9 = -5 = g(3)$

$g(-1) = 4-(-1)^2$

$g(-1) = 4-1 = 3$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$(-1, 3) \; (3, -5) \quad \frac{-8}{4} = \frac{-2}{1}$

$y-(3) = -2(x-(-1))$

$y - 3 = -2x - 2$

$+3$

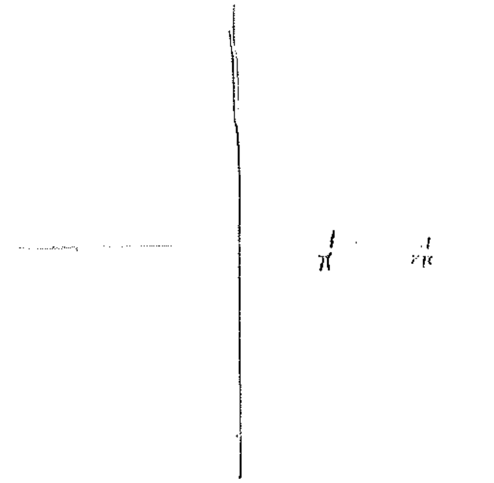
$y = -2x + 1$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.

$f(x)$

Amp



$f(x) = -4\sin(x)$
$A = 4$

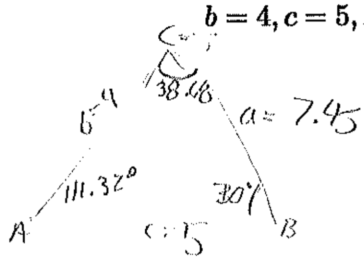
Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

Name ______________ **ID#** ______________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$b = 4, c = 5, B = 30°$

A: 111.32°
B: 30°
C: 38.68
a: 7.45
b: 4
c: 5

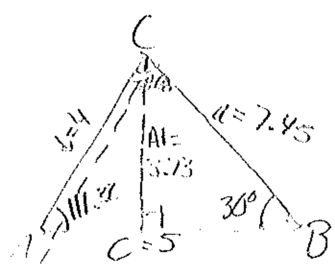


$\frac{\sin 111.32°}{a} = \frac{\sin 30°}{4}$

$\frac{a \cdot \sin 30}{\sin 30} = \frac{4 \cdot \sin 111.32°}{\sin 30} =$

$\frac{\sin 30°}{1} = \frac{A}{7.45}$

$7.45 \cdot \sin 30 = AH\ 3.73$



3.73 < 7.45 > 4

One triangle

$\frac{\sin 30°}{4} = \frac{\sin C}{5}$

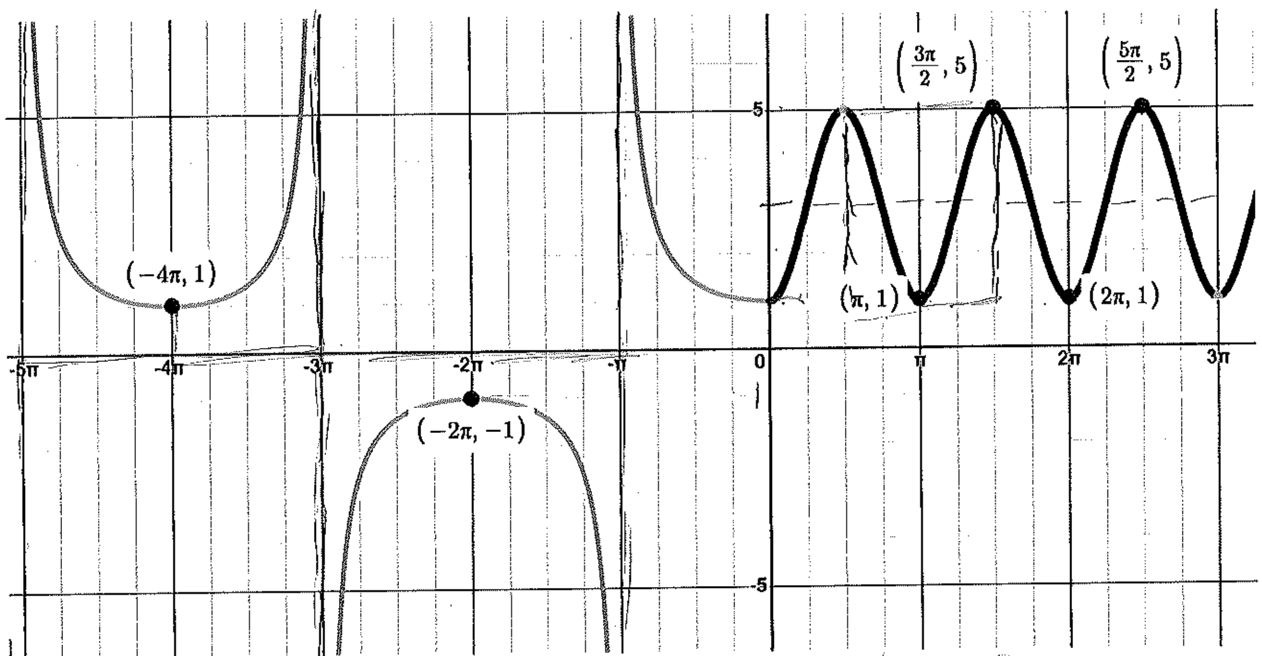
$\frac{5 \cdot \sin 30°}{4} = \frac{4 \cdot \sin C}{4}$

$\frac{5 \cdot \sin 30°}{4} = 0.625$

$\sin^{-1}(0.625) = 38.68° = C$

$180° - (38.68° + 30°) = 111.32° = A$

5) Given the graph below find the function $f(x)$ and state the domain and range.



secant

$f(x) = 2\sec(2\pi x)$

Period: 2π

Amp

$1-(-1) = 2$ Amp $= 2$

$\frac{}{1} = 2$

$-1(-1)$

0

domain: All real numbers except by a factor -2π

range: $(-\infty, -1] \cup [1, \infty)$

$A\cos(wx) + B$

$f(x) = 2\cos(2x) + 3$

$5 - 1 = \frac{4}{2} = 2$ Amp

Period: $\frac{2\pi}{\pi} = 2$

$B: 5+1 = 6$ 3

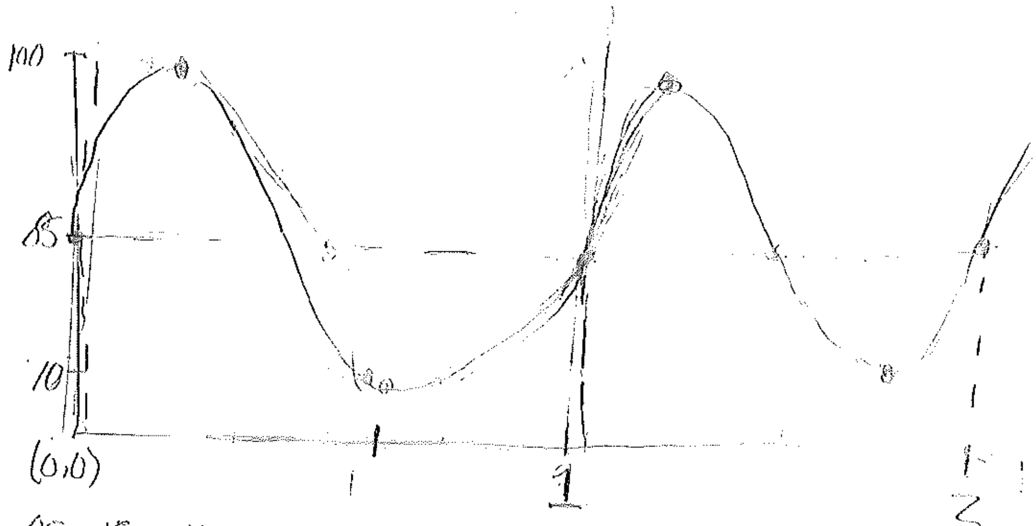
domain: $(1, \infty)$

range: $(1, 5)$

Name _______________ ID# _______________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



$A = 85 + 15 = 100$

$85 - 15 = 70$

$B: 2\pi$

period $= \frac{2\pi}{2\pi} = 1$ sec

Midline: 85

b) The systolic pressure is ___100___

c) The diastolic pressure is ___70___

d) The heart rate is ___60___ beats per minute

Beat in a sec = 1

Name [redacted] **ID#** [redacted]

Math 104 Spring 2025 Final Exam

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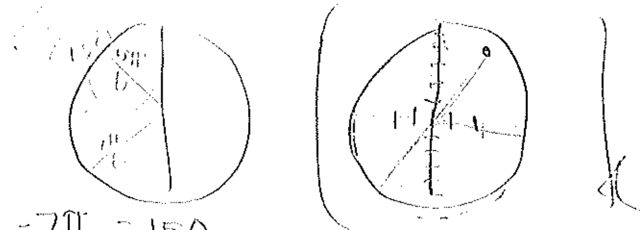
Total		19 /30



$x = r\cos(\theta)$

$y = r\sin(\theta)$

$\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

$$\begin{array}{cc} x & y \\ \end{array}$$
$$\left(-\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$$

$-\frac{7\pi}{6} = 150$

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

$r = 5 \qquad \theta = \frac{-7\pi}{6}$

$y = 5\sin\left(\frac{-7\pi}{6}\right)$

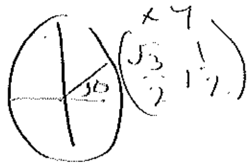
$x = 5\cos\left(\frac{-7\pi}{6}\right) \qquad x = 5\cdot\left(-\frac{\sqrt{3}}{2}\right)$

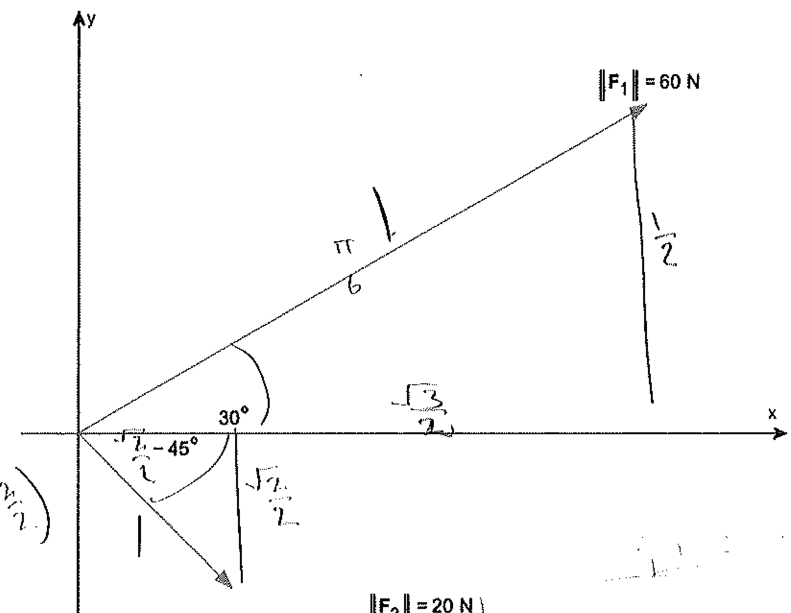
$\sin\left(\frac{-7\pi}{6}\right) = \sin(150) = \frac{1}{2}$

$\cos(150) = -\frac{\sqrt{3}}{2} \qquad x = -\frac{5\sqrt{3}}{2}$

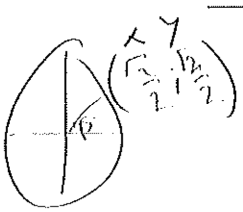
$y = 5\cdot\left(\frac{1}{2}\right) = \frac{5}{2}$

1b) Add the two vectors $F_1 + F_2$





	y	x
$1(60)$	$\frac{1}{2}(60)$	$\frac{\sqrt{3}}{2}(60)$
60	30	$30\sqrt{3}$



$F_1 + F_2 = 30\sqrt{3} + 10\sqrt{2}\,\hat{i} + 30 + 10\sqrt{2}$

	x	y
$1(20)$	$\frac{\sqrt{2}}{2}(20)$	$\frac{\sqrt{2}}{2}(20)$
20	$10\sqrt{2}$	$10\cdot 10$

Name ______________________ ID# ______________________

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

1. $4-(-1)^2 = 3$ 2. $4-(3)^2 = -5$

ROC $= \frac{-5-3}{3-(-1)} = \boxed{-1.67}$

Point 1: $(-1,3)$

point 2: $(3,-5)$

b) Find an equation of the secant line containing $(\overset{x_1}{-1}, \overset{gx_1}{g(-1)})$ and $(\overset{x_2}{3}, \overset{gx_2}{g(3)})$.

$y-3 = -1.67(x-(-1))$

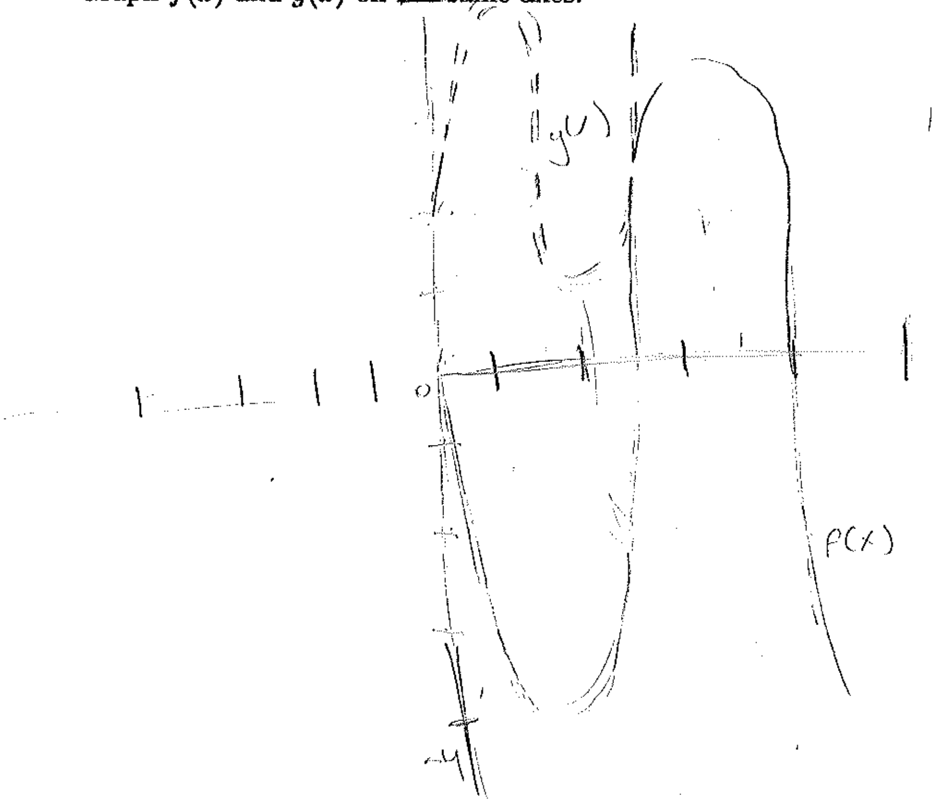
$y-3 = -1.67x + 1.67$

$+3 \qquad +3$

$\boxed{y = -1.67x + 4.67}$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$-4\sin(x) = 2\sin(x) + 3$

$+4\sin(x) \qquad +4\sin(x)$

$0 = 2\sin(x) + 3 + 4\sin(x)$

$0 = 2x + 3 + 4x$

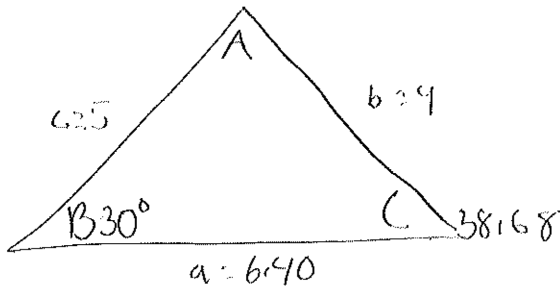
$0 = |6x + 3|$

Name ____________________ ID# ____________________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$b = 4, c = 5, B = 30°$

two triangles



$A_1 = 111.32$ $a = 6.46$

$B_1 = 30$ $b_1 = 4$

$C_1 = 38.68$ $c_1 = 5$

$a^2 = b^2 + c^2$

$a^2 = 4^2 + 5^2$

$a^2 = 16 + 25$

$\cancel{a^2} = 41$

$\underline{a = 6.40}$

$\frac{\sin C}{5} \times \frac{\sin 30}{4}$

$\frac{5 \sin 30}{4} = \frac{\cancel{4} \sin(C)}{\cancel{4}}$

$\frac{5 \sin 30}{\sin^{-1} 4} = \frac{\cancel{\sin}(C)}{\sin^{-1}}$

$\sin^{-1}\left(\frac{5\sin(30)}{4}\right)$

$\underline{C = 38.68}$

A

$180 - 38.68 - 30 = \underline{111.32}$

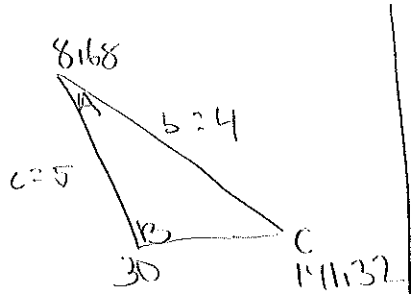
$180 - 38.68 = 141.32$

C B

$141.32 + 30 + \underline{\quad} = 180$

A

8.68



$A_2 = 8.68$

$B_2 = 30$

$C_2 = 141.32$

$a_2 = 1.92$

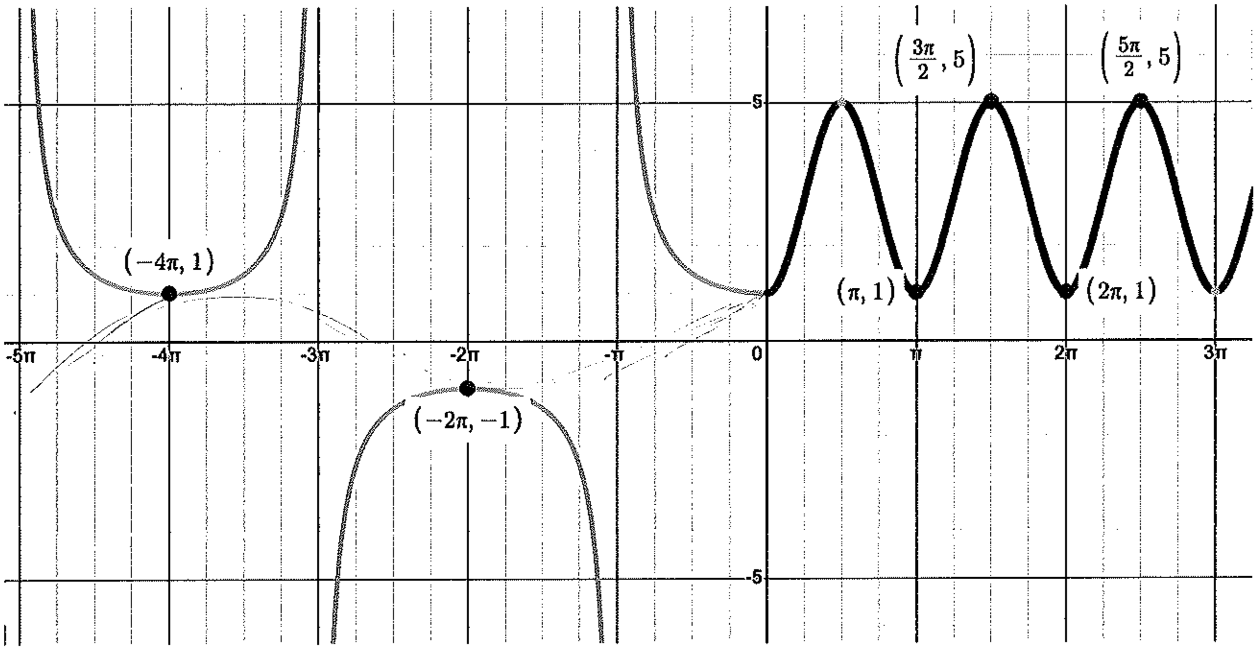
$b_2 = 4$

$c_2 = 5$

$\frac{\sin 23.13}{5} = \frac{\sin(8.68)}{a}$

$\frac{5\sin(8.68)}{\sin(23.13)} = \underline{1.92}$

5) Given the graph below find the function $f(x)$ and state the domain and range.



Red graph : Domain =

Range = $(-\infty, -1] \cup [1, \infty)$

black graph: Domain = $[1, \infty)$

Range = $[1, 5]$

Name ______________________ ID# ______ ______________________

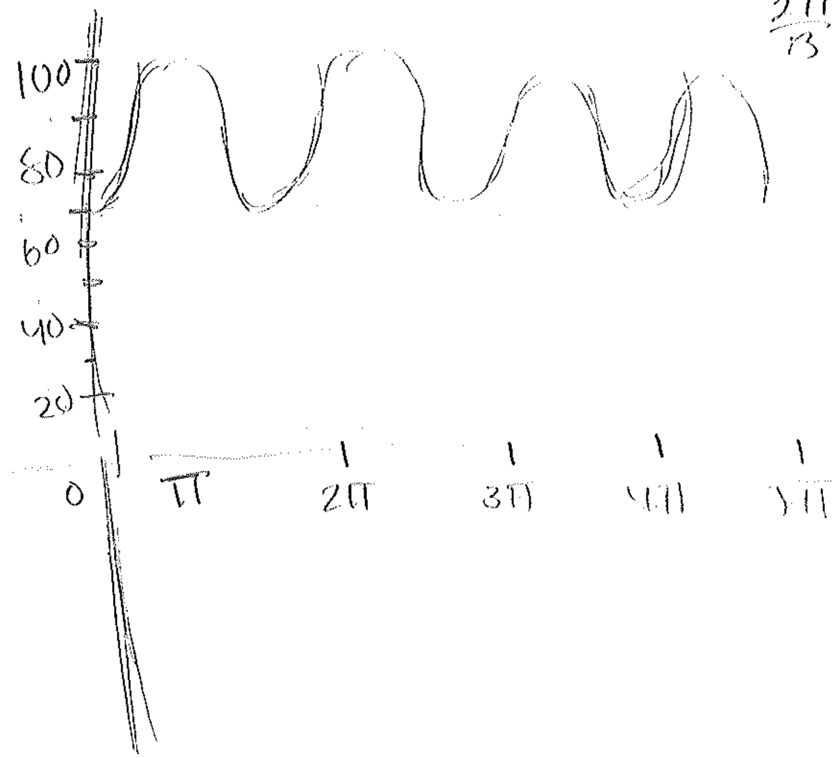
6) Blood pressure is modeled by the function $P(t) = \overset{A}{15}\sin(\overset{B}{2\pi} t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

period = 2π

Amp = 15

a) Graph $P(t)$ for two periods, label your axes and local maximums.

$\frac{2\pi}{B} = \frac{2\pi}{2\pi} = 1$



60 secs
1 period = 60

15 + 85 = 100

b) The systolic pressure is 100

c) The diastolic pressure is 70 ← 85 − 15 = 70

d) The heart rate is 60 beats per minute

Name [redacted] ID# [redacted]

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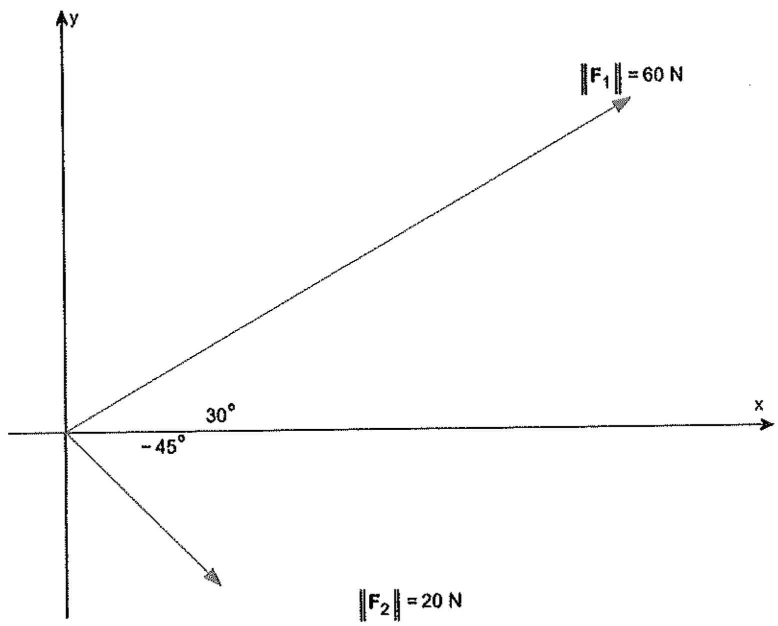
Total		19 /30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

$x = -4.33$

$y = 2.5$

1b) Add the two vectors $F_1 + F_2$



Name Leo Ahumaya ID# 5550209476

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from $\overset{a}{-1}$ to $\overset{b}{3}$.

$g(-1) = 4 - (-1)^2$

$4 - (1)$

3

$g(3) = 4 - (3)^2$

$4 - 9$

-5

$\frac{f(b) - f(a)}{b - a} = m$

$\frac{-5 - 3}{3 - (-1)} = \frac{-8}{4} = \boxed{-2}$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$(-1, 3) \quad (3, -5)$

$y - y_1 = m(x - x_1)$

$y - 3 = -2(x + 1)$

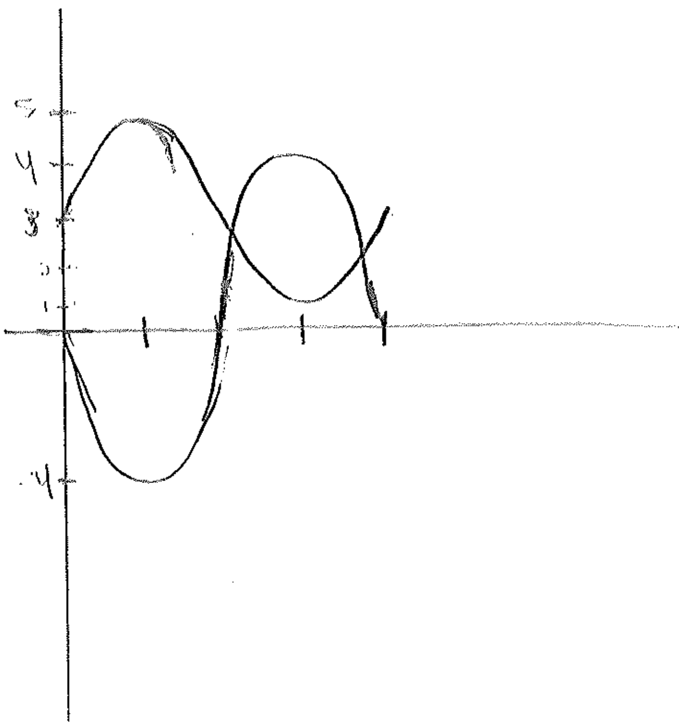
$y - 3 = -2x - 2$

$+3 \qquad +3$

$\boxed{y = -2x + 1}$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

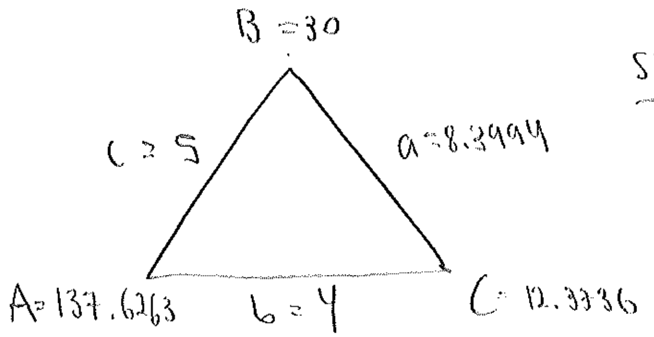
$-4\sin(x) = 2\sin(x) + 3$

$0 = 4\sin(x) + 2\sin(x) + 3$

Name Leo Ahumada ID# 555 0209476

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30°$$



$$\frac{\sin(30)}{4} = \frac{\sin(C)}{5}$$

$$\sin(C) = \frac{5 \cdot \sin(30)}{4}$$

$$\sin^{-1}\left(\frac{5 \cdot \sin(30)}{4}\right) = \underline{C}$$

$$12.3736 = \underline{C}$$

$$180 - (30 + 12.3736) = 137.6263$$

$$a^2 = b^2 + c^2 - 2bc \cos(A)$$

$$a^2 = 4^2 + 5^2 - 2(4)(5)\cos(137.6263)$$

$$\sqrt{a^2} = \sqrt{70.55}$$

$$a = 8.3994$$

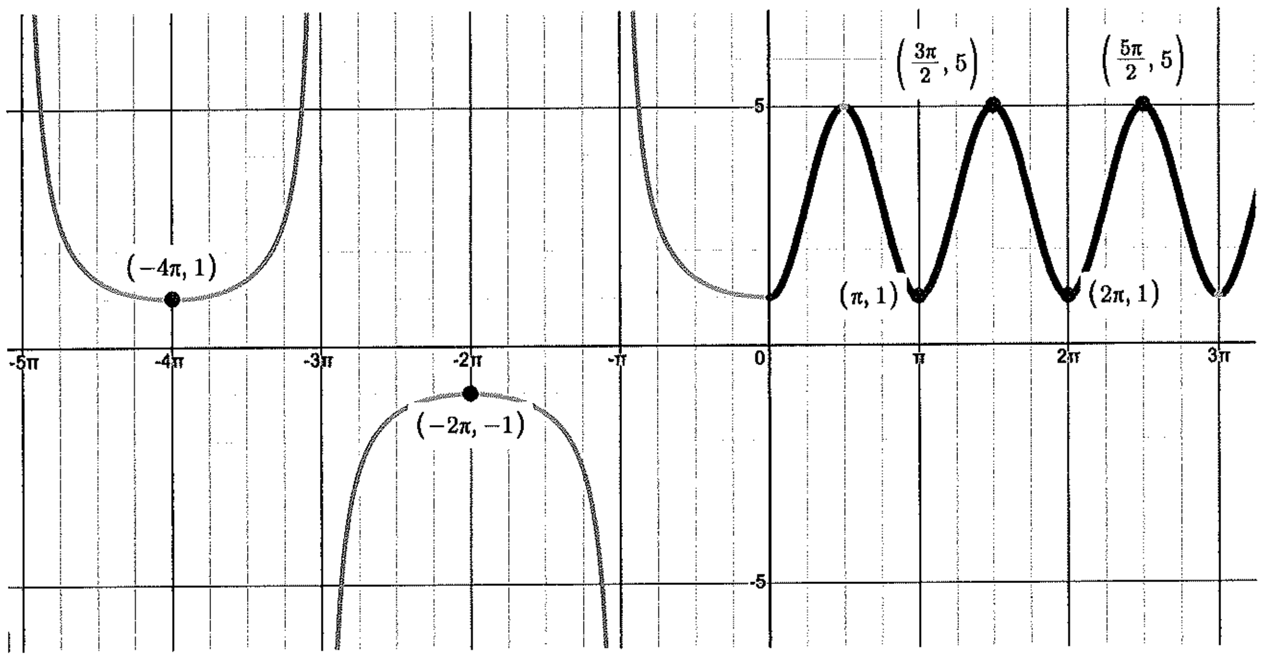
Side opposite the largest angle is largest side: TRUE (possible triangle)

Possible second triangle: $B_2 = 180 - C_1 = 180 - 12.3736 = 167.6264$

$$\begin{array}{r} 167.6264 \\ +\ 30 \\ \hline 197.6264 > 180 \end{array}$$

$\boxed{\text{NO 2nd triangle}}$

5) Given the graph below find the function $f(x)$ and state the domain and range.



domain: $(-\infty, \infty) \cap (-\pi \cdot |2\pi k|)$

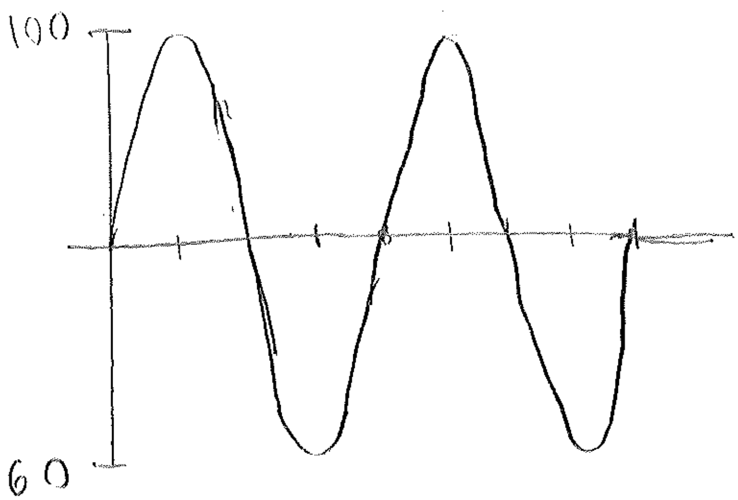
range: $(-\infty, -1] \cup [1, \infty)$

$\angle 0$:

Name ______________________ **ID#** ______________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



Period $= \frac{2\pi}{2\pi} = 1$

bpm $= \frac{60}{1} = 60$

b) The systolic pressure is 100

c) The diastolic pressure is 60

d) The heart rate is 60 beats per minute

Name [redacted] ID# [redacted]

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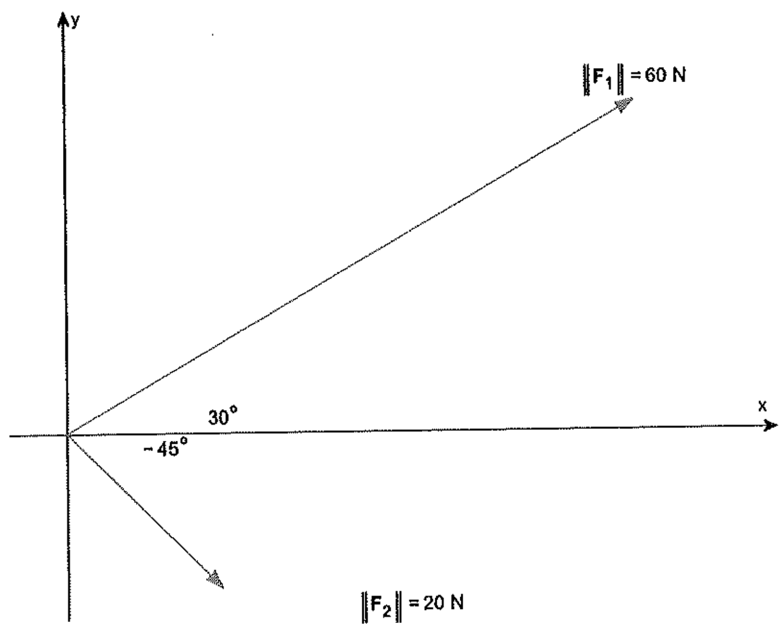
Total		/30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

1b) Add the two vectors $F_1 + F_2$



Name ______________________ **ID#** ______________________

2) Let $g(x) = 4 - x^2$.
Suspicious, Never did a problem with a negative

a) Find the average rate of change from -1 to 3.

Average rate of change -1 to 3

$g(3) = 3^2 + 4 = 13$

$g(-1) = -1^2 + 4 = 5$

$\frac{13-5}{3-1} = \frac{8}{4} = 2$

$g(3) = -3^2 + 4 = 13$

$g(-1) = (1)^2 + 4 = 5 = y_1$

$\frac{13-5}{3+1} = \frac{8}{4} = 2 = m$

Substitute $m = 2$; $(x_1, y_1) = (-1, 5)$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

Equation of secant line containing $(1, g(-1))$ and $(3, g(3))$

$y - 5 = 2(x + 1) = \boxed{2(x+1)}$

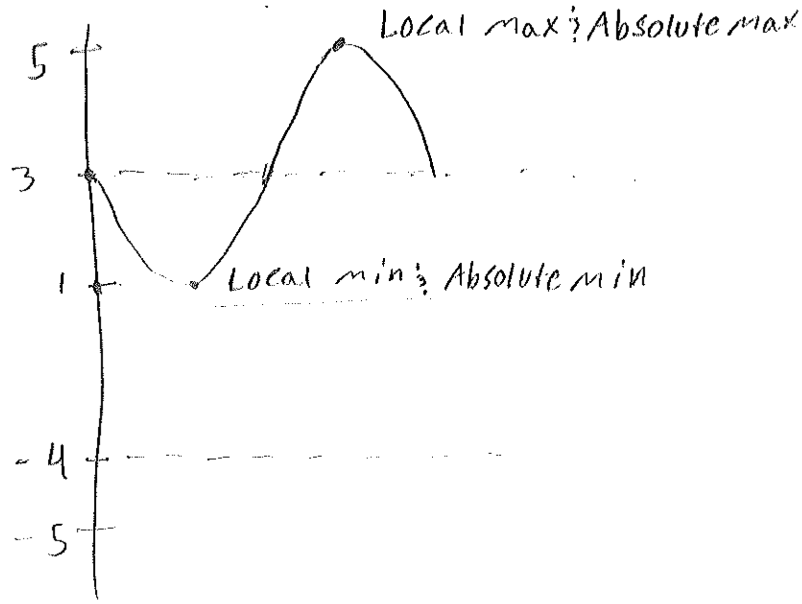
Slope intercept form:

$y = 2(x+1) + 5 = 2x + 2 + 5 = \boxed{y = 2x + 7}$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

$3+2=5$
$3-2=1$

Graph $f(x)$ and $g(x)$ on the same axes.



$1-4=3$
$1+4=5$

Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

Name ______________________ **ID#** ______________________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$$b = 4, c = 5, B = 30°$$

$B = 30°, b = 4, c = 5$

$\frac{\sin B}{b} = \frac{\sin C}{c} \Rightarrow \frac{\sin 30°}{3} = \frac{\sin C}{5}$

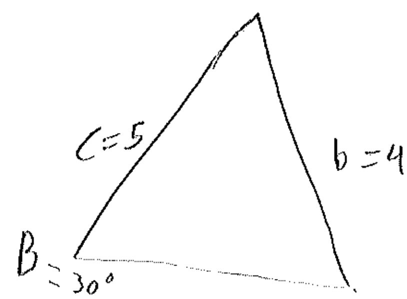
Two Triangles

$\frac{5 \cdot \sin 30°}{3} = \boxed{56.44}$ C_1

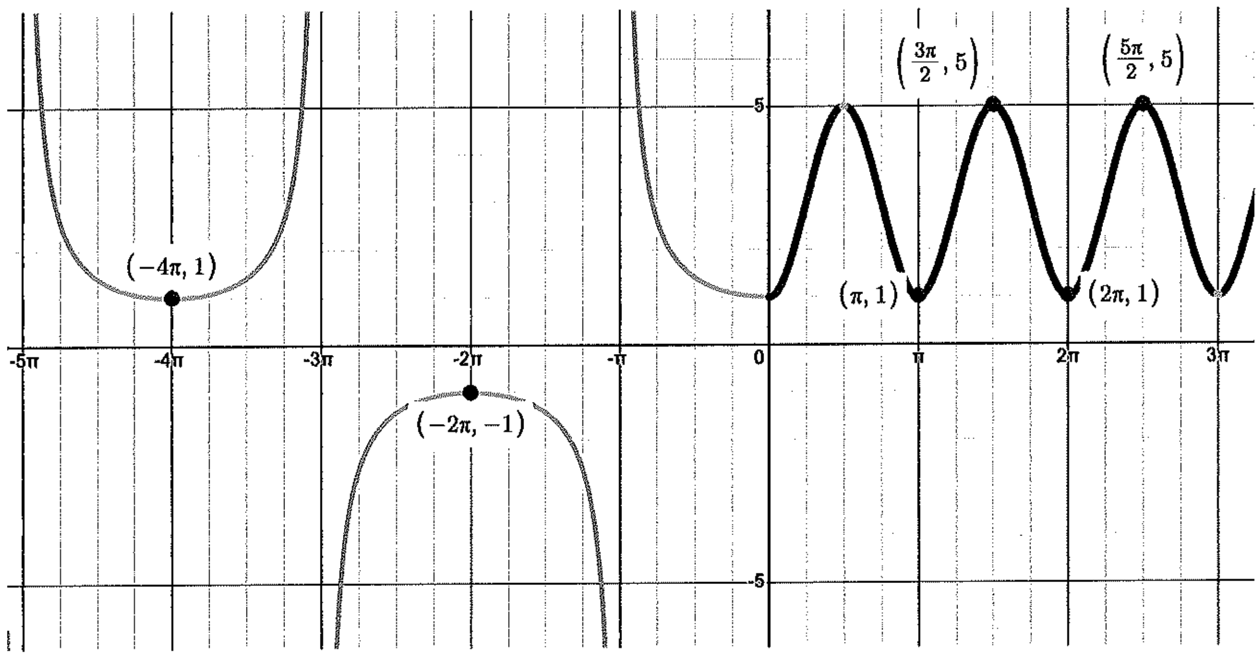
56.44269024

$180 - 56.44 - 30 = \boxed{93.56}$ A_1

$180 - 56.44269024 = \boxed{123.5573098}$ C_2



5) Given the graph below find the function $f(x)$ and state the domain and range.

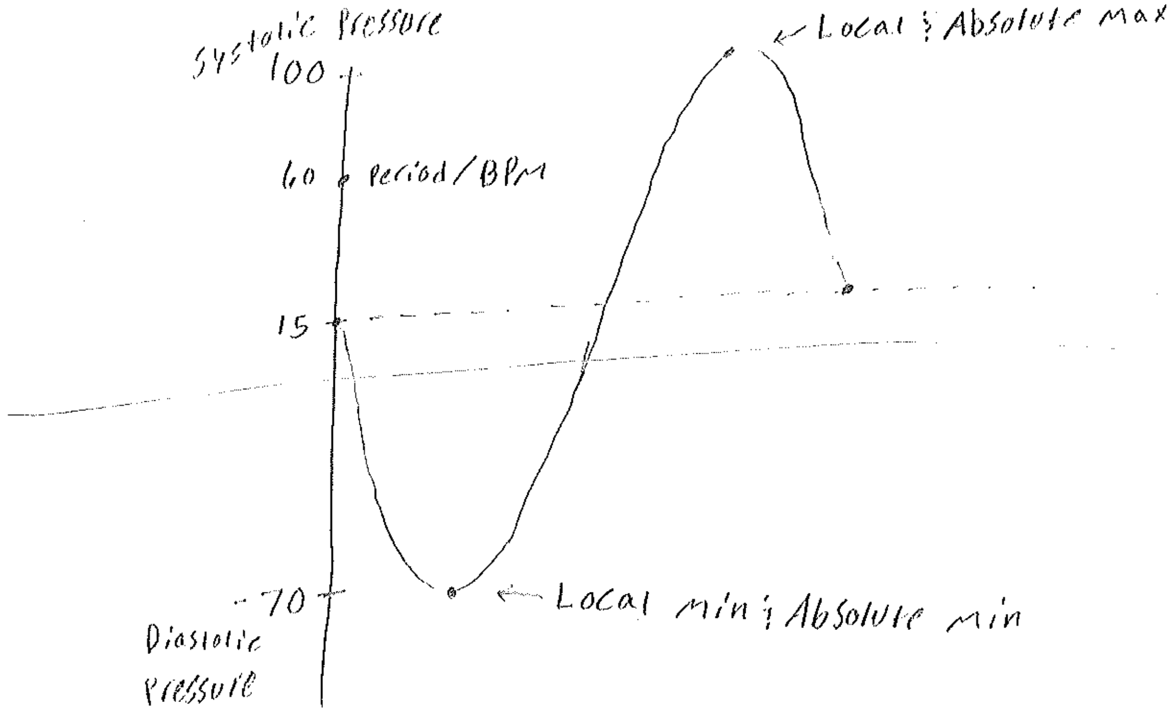


Name ______________________ ID# ______________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

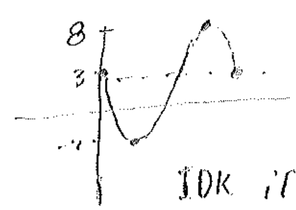
a) Graph $P(t)$ for two periods, label your axes and local maximums.

$15 + 85 = 100$
$15 - 85 = -70$



Ex This morning = $Y = 5\sin x + 3$

$3 + 5 = 8$
$3 - 5 = -2$



IDK if this is the same thing

b) The systolic pressure is <u>100</u>

c) The diastolic pressure is <u>70</u>

d) The heart rate is <u>60</u> beats per minute

$P(t) = 15\sin(2\pi t) + 85$

Systolic Pressure $= 15 + 85 = \boxed{100}$

Dystolic Pressure $= -15 + 85 = \boxed{70}$

Period for Heart rate

$\frac{2\pi}{2\pi} = 1$ second $\qquad$ BPM $= \frac{60}{T} = \frac{60}{1} = \boxed{60 \text{ BPM}}$

Name [redacted] **ID#** [redacted]

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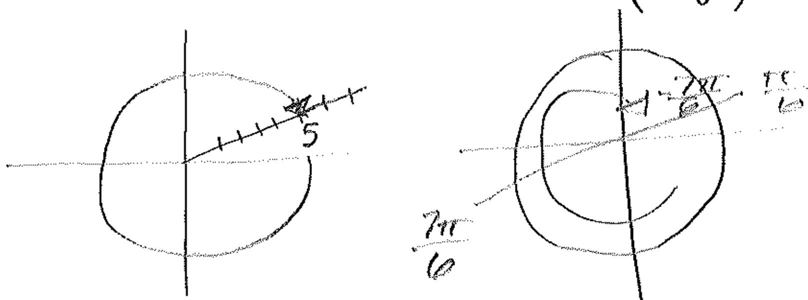
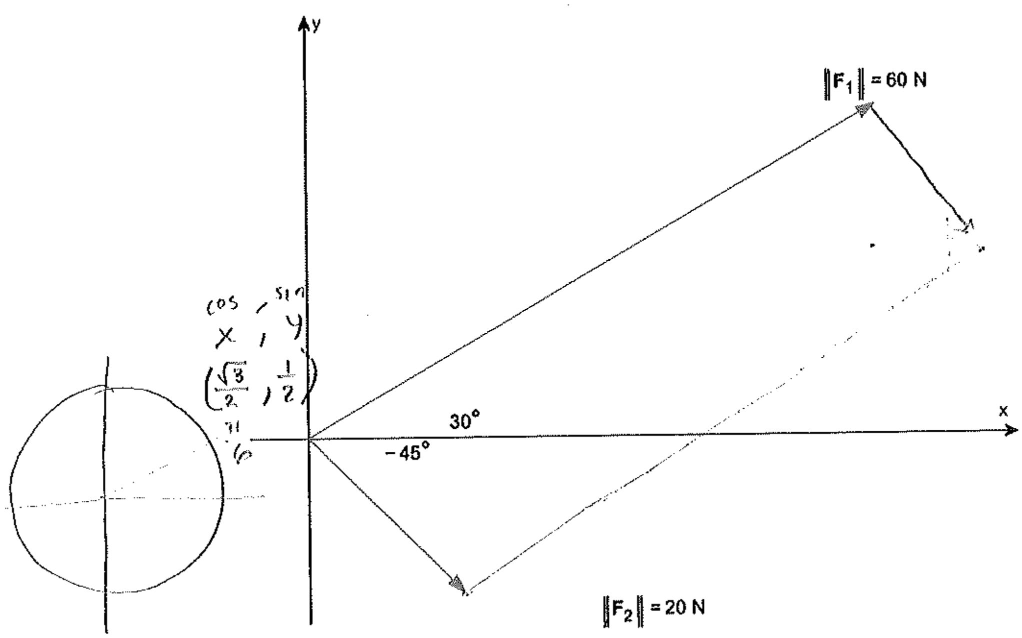
1a) Plot the point is given in polar coordinates $(r,\theta) = \left(5, \frac{-7\pi}{6}\right)$



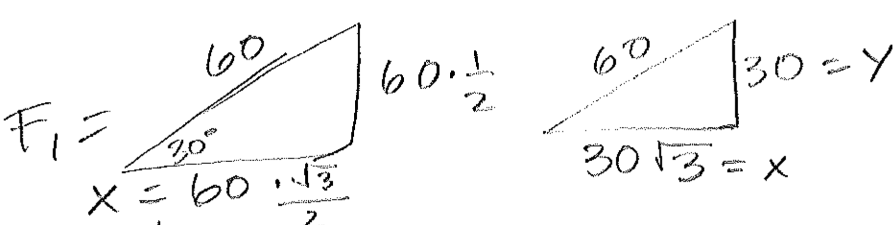
1b) Add the two vectors $F_1 + F_2 = (\quad)\hat{i} + (\quad)\hat{j}$

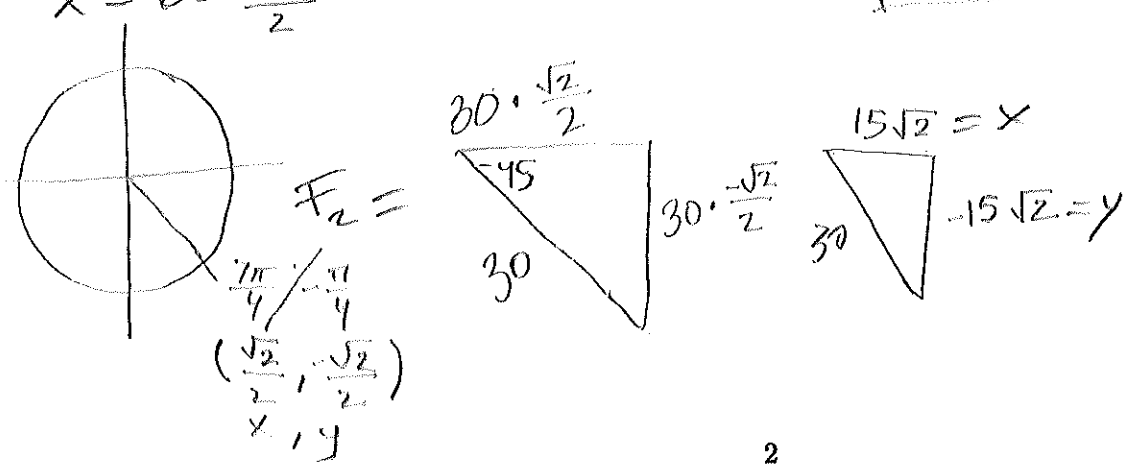


Horizontal $(\hat{i})$
$30\sqrt{3} + 15\sqrt{2}$

vertical $(\hat{j})$
$30 - 15\sqrt{2}$

$\boxed{(30\sqrt{3} + 15\sqrt{2})\hat{i} + (30 - 15\sqrt{2})\hat{j}}$





Name ________ ID# ________

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

$(-1, \underline{3})$

$g(\ ) = 4 - (\ )^2$

$g(-1) = 4 - (-1)^2$

$= 4 - 1$

$= 3$

$(3, \underline{-5})$

$g(\ ) = 4 - (\ )^2$

$g(3) = 4 - (3)^2$

$= 4 - 9$

$= -5$

$m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{-5 - 3}{3 - (-1)} = \dfrac{-8}{3+1} = -\dfrac{8}{4} = -2$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

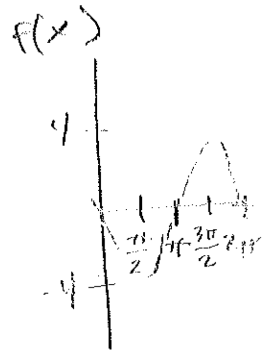
$y - y_1 = m(x - x_1)$

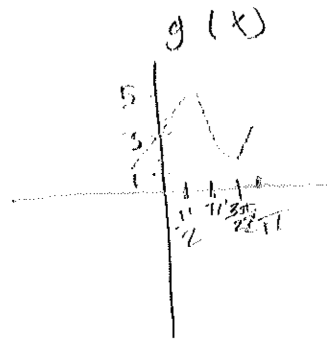
$y - 3 = -2(x + 1)$

$y - 3 = -2x - 2$

$+3 \qquad +3$

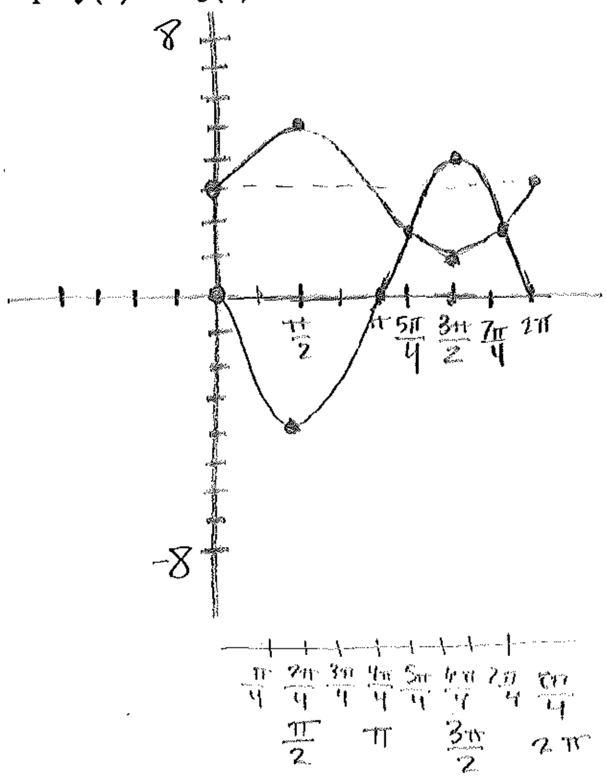
$y = -2x + 1$





3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.

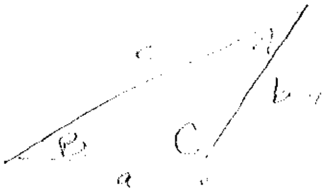


Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$f(x) = g(x) \rightarrow \frac{5\pi}{4}, \frac{7\pi}{4}$

Nan. ID#

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).



$b = 4, c = 5, B = 30°$ find $a, C, \& A$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin A}{a} = \frac{\sin 30}{4} = \frac{\sin C}{5}$$

171.32

$$\frac{\sin 30}{4} = \frac{\sin C}{5}$$

$$\frac{4\sin C}{4} = \frac{5\sin 30}{4}$$

$$\sin C = \frac{5\sin 30}{4}$$

$$C = \sin^{-1}\left(\frac{5\sin 30}{4}\right)$$

$$C = \sin^{-1}\left(\frac{5(0.5)}{4}\right)$$

$$C = \sin^{-1}\left(\frac{2.5}{4}\right)$$

$$C = \sin^{-1}(0.625)$$

68.68 $C \approx 38.682187$

$\boxed{C \approx 38.68}$

$180 - B - C = \boxed{111.32 = A}$

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin 111.32}{a} = \frac{\sin 30}{4}$$

$$\frac{a\sin 30}{\sin 30} = \frac{4\sin 111.32}{\sin 30}$$

$$a = \frac{4\sin 111.32}{\sin 30}$$

$$a = \frac{4(.9315647)}{(.5)}$$

$$a \approx \frac{3.726256}{.5}$$

$a \approx 7.452512$

$\boxed{a \approx 7.45}$

possible 2Δ:

$180 - C_1 = C_2$

$180 - 38.68 = \boxed{141.32 = C_2}$

$180 - B - C_2 = A_2$

$180 - 30 - 141.32 =$

$180 - 171.32 =$

$\boxed{8.68 = A_2}$

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 8.68}{a} = \frac{\sin 30}{4}$$

$$\frac{a\sin 30}{\sin 30} = \frac{4\sin 8.68}{\sin 30}$$

$$a = \frac{4\sin 8.68}{\sin 30}$$

$$a = \frac{4(\quad)}{(\quad)}$$

$$a \approx \frac{4(0.15091562)}{(0.5)}$$

$$a \approx \frac{0.60366248}{0.5}$$

$a \approx 1.20732496$

$\boxed{a \approx 1.21}$

5) Given the graph below find the function $f(x)$ and state the domain and range.

$A = \frac{5-1}{2} = 2$

$M = \frac{5+1}{2} = 3$

$\frac{2\pi}{B} = 4\pi$

$2\pi = 4\pi B$

$\frac{1}{2} = B$

$\sec(\frac{1}{2}x)$

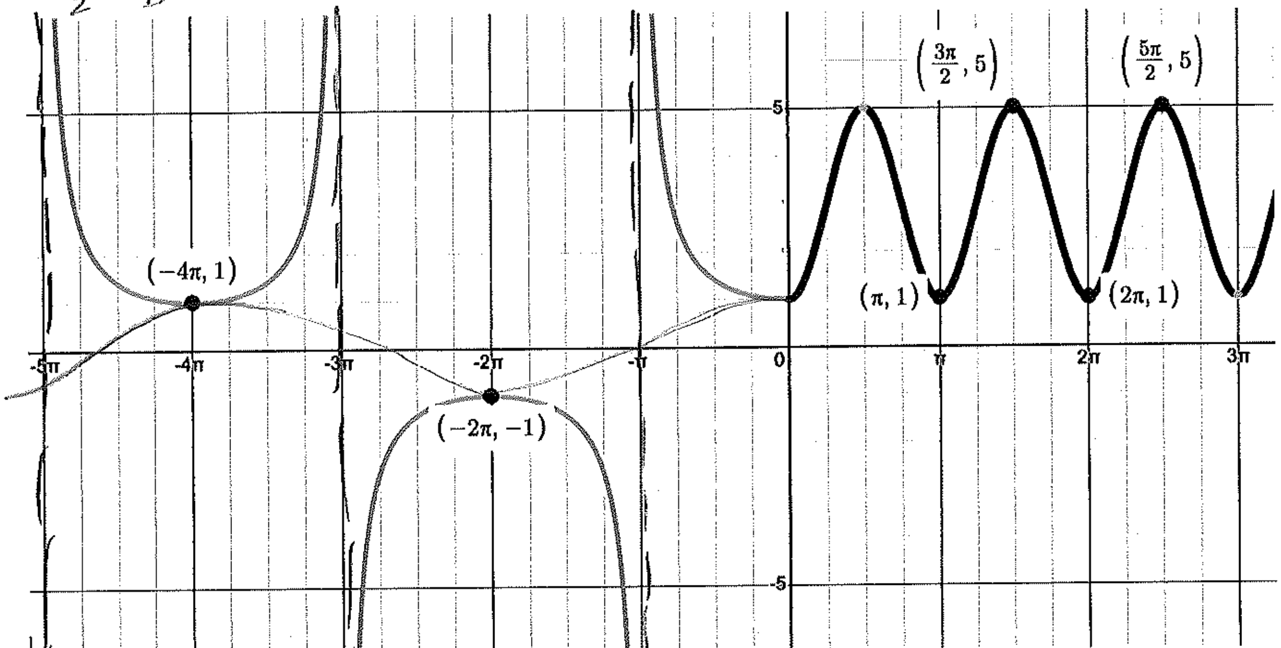
$\frac{2\pi}{B} = \pi$

$2\pi = \pi B$

$\frac{2\pi}{\pi} = B$

$2 = B$

$-2\cos(2x)+3$



$f(x) = \sec(\frac{1}{2}x) - 2\cos(2x) + 3$

$D = x \mid x \neq k\pi$

$k = 3, 5, 7 \ldots$

$D = (0, \infty)$

$D = [-5\pi, -3\pi] \cup [-3\pi, -\pi] \cup [-\pi, \infty]$

$R = (-\infty, \infty) \cup [1, 5]$

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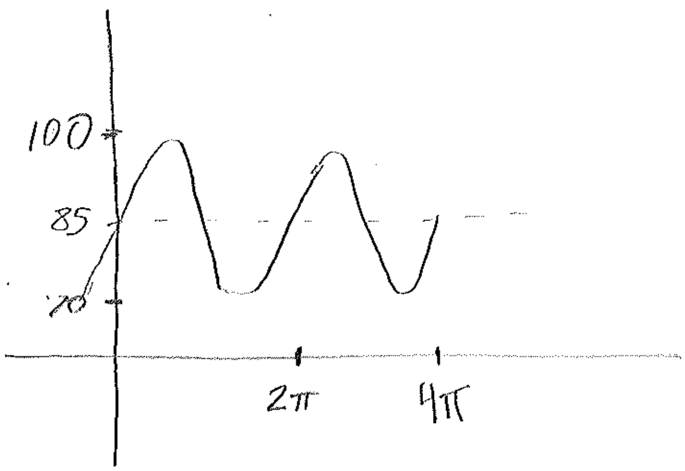
6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



$15(\sin(2\pi t)+85$ $\qquad$ $15(\sin(2\pi t)+85$

$15(\quad)+85$ $\qquad$ $15(\quad)+85$

$15(1)+85$ $\qquad$ $15(-1)+85$

100 $\qquad$ 70

$\frac{2\pi}{2\pi} = 1 = T$ $\quad HR = \frac{60}{T} = \frac{60}{1} = 60$

b) The systolic pressure is ___100___

c) The diastolic pressure is ___70___

d) The heart rate is ___60___ beats per minute

Name [redacted] ID# [redacted]

Math 104 Spring 2025 Final Exam

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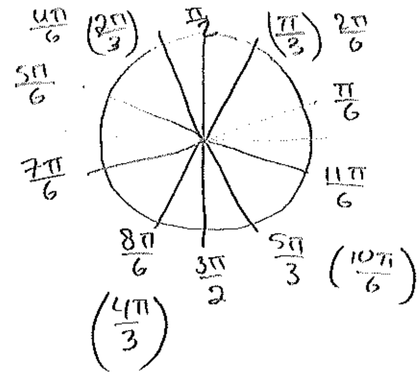
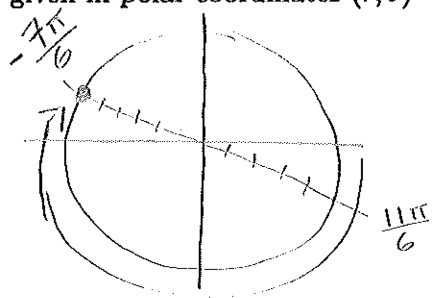
Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		28.5/30

Unit Circle



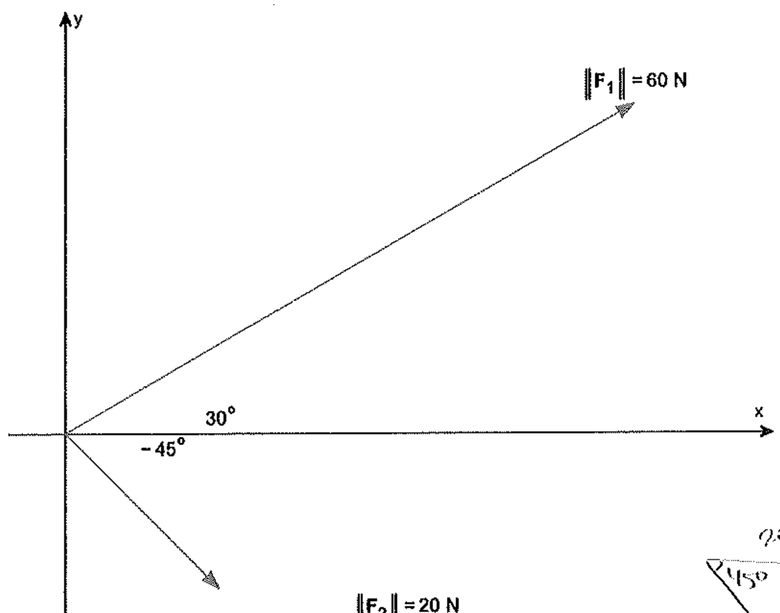
1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$

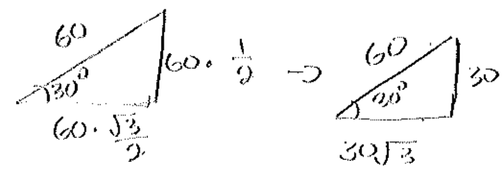
$\left(5, \frac{-7\pi}{6}\right)$

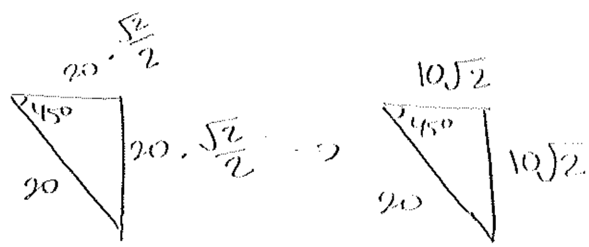


1b) Add the two vectors $F_1 + F_2$

$F_1 = 60N$ $F_2 = 20N$







$\boxed{(30\sqrt{3} + 10\sqrt{2})i}$

$\boxed{(30 - 10\sqrt{2})j}$

Name ______________________ ID# ______________________

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

$g(-1) = 4 - (-1)^2$

$= 4 - 1 = 3$

y_1 $\boxed{g(-1) = 3}$

$\frac{f(b) - f(a)}{b - a}$

$\boxed{\frac{y_2 - y_1}{x_2 - x_1}}$

$g(3) = 4 - (3)^2$

$4 - 9 = -5$

y_2 $\boxed{g(3) = -5}$

$\frac{-5 - 3}{3 - (-1)} = \frac{-8}{4} = \boxed{-2}$ rate of change

b) Find an equation of the secant line containing $(\overset{x_1}{-1}, \overset{y_1}{g(-1)})$ and $(\overset{x_2}{3}, \overset{y_2}{g(3)})$.

$y - y_1 = m(x - x_1)$

$y - (3) = -2(x - (-1)) \rightarrow y - 3 = -2(x + 1)$

$= y - 3 = -2x - 2$

$+3 \qquad +3$

$\boxed{y = -2x + 1}$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

$f(x)$ $\quad$ $g(x)$

$-4\sin(x)$ $\quad$ $2\sin(x)+3$

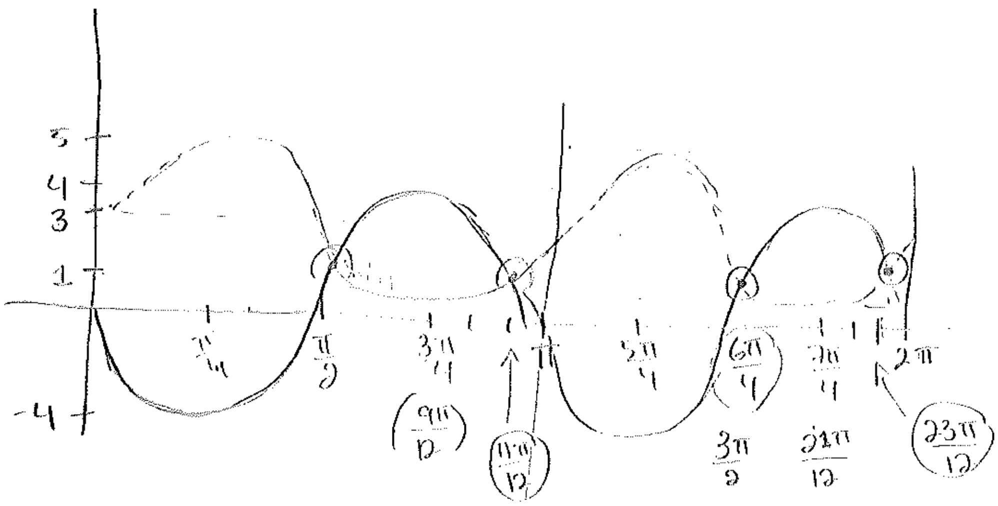
$2+3=5$

$-2+3=1$ Amplitudes

$\frac{5+1}{2} = \frac{6}{2} = 3$

Midline

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$f(x) = g(x)$ on the points:

$$\left(\frac{\pi}{2}, \frac{11\pi}{12}, \frac{3\pi}{2}, \frac{23\pi}{12}\right)$$

correct for 2 ✗

Name ________________________ ID# ________________________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$\boxed{S.S.A}$

$b = 4, c = 5, B = 30°$

$$\frac{\sin 30}{4} = \frac{\sin C}{5}$$

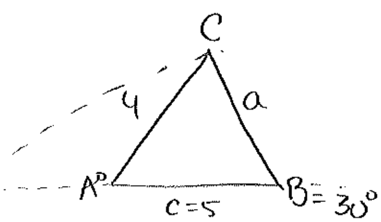
$$\frac{\sin 30 \cdot 5}{4} = \frac{\sin C \cdot 4}{4}$$

$$\sin C = \frac{5 \sin 30}{4}$$

$\boxed{C° = 38.68}$
Small Angle $C°$

$$\sin^{-1} C = 38.6822$$

$$180 - (30 + 38.6822) = \boxed{111.3178}$$
Small Angle $A°$



$$\frac{\sin 30}{4} = \frac{\sin 111.3178}{a}$$

$$\frac{\sin 30 \cdot a}{\sin 30} = \frac{4 \cdot \sin 111.3178}{\sin 30}$$

$$\boxed{\text{Small Side } a} = \frac{4 \sin 111.3178}{\sin 30}$$

$$= \boxed{7.4526}$$

Big $C°$ Angle

$$180 - 38.6822 = \boxed{141.3178}$$

$$180 - (30 + 141.3178)$$

$$= \boxed{8.6822}$$
Big Angle $A°$

$$\frac{\sin 30}{4} = \frac{\sin 8.6822}{a}$$

$$\frac{\sin 30 \cdot a}{\sin 30} = \frac{\sin 8.6822 \cdot 4}{\sin 30}$$

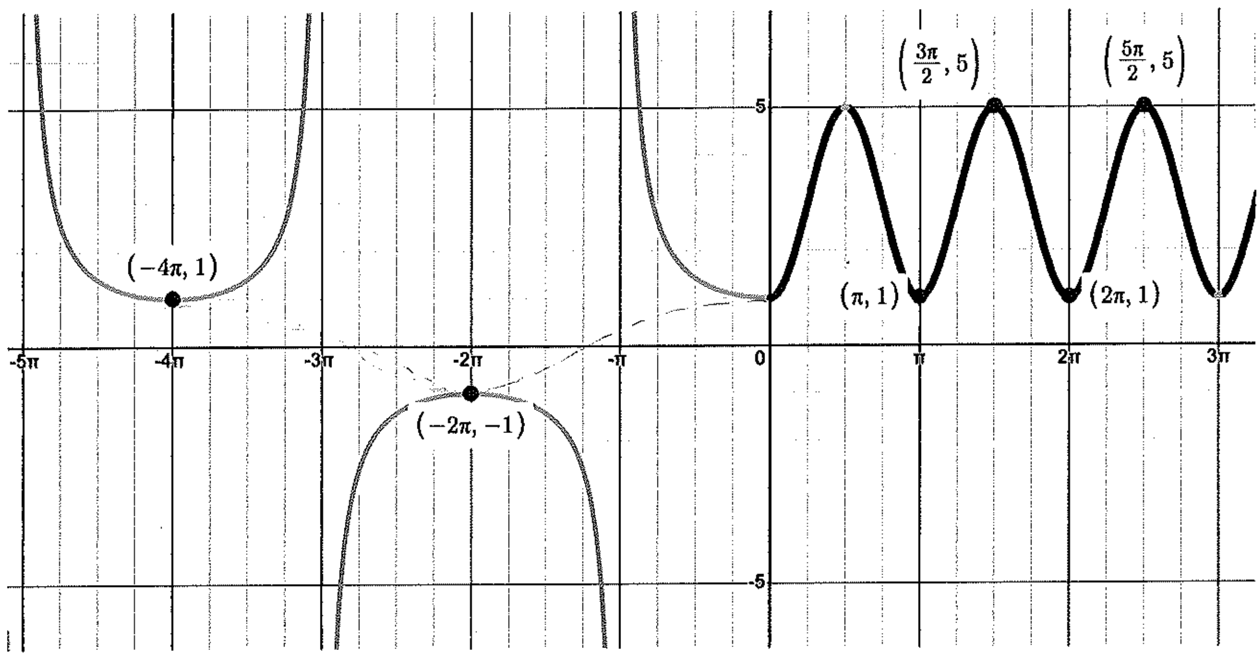
$$a = \frac{4 \sin 8.6822}{\sin 30}$$

$\boxed{\text{Big side } a}$

$$= \boxed{1.2076}$$

There are 2 possible triangles

5) Given the graph below find the function $f(x)$ and state the domain and range.



Period

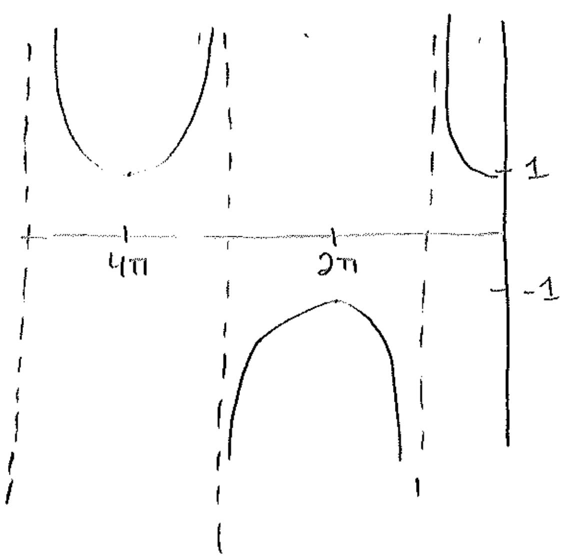
$\frac{2\pi}{4\pi} = \frac{1}{2}$

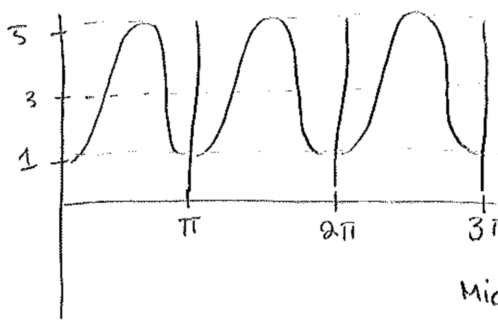
$\boxed{\sec\left(\frac{1}{2}x\right)}$

Period $= \frac{2\pi}{\pi} = 2$

Amplitude

$\frac{5-1}{2} = 2$





$-2+3=1$

$2+3=5$

Midline

$= \frac{5+1}{2} = 3$

$\boxed{-2\cos(2x)+3}$

Domain $=$ x is all $\mathbb{R}$ except $-\pi + k$ where k is multiples of -2π, $[0,\infty)$

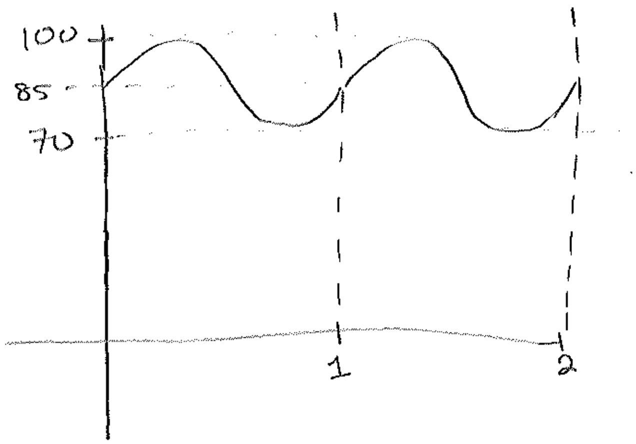
Range $= (-\infty,-1] \cup [1,5] \cup [5,\infty)$

Name ______________________ ID# ______________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

$15\sin(2\pi t) + 85$

a) Graph $P(t)$ for two periods, label your axes and local maximums.



$15 + 85 = 100$ $\quad \frac{170}{2} = 85$

$-15 + 85 = 70$ $\quad$ Midline

Period $= \frac{2\pi}{2\pi} = \boxed{1}$

Heart Rate $= \frac{60}{1} = 60$

b) The systolic pressure is 100

c) The diastolic pressure is 70

d) The heart rate is 60 beats per minute

Nam[redacted] **ID#** [redacted]

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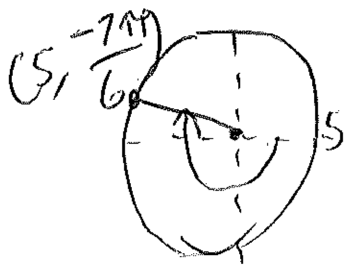
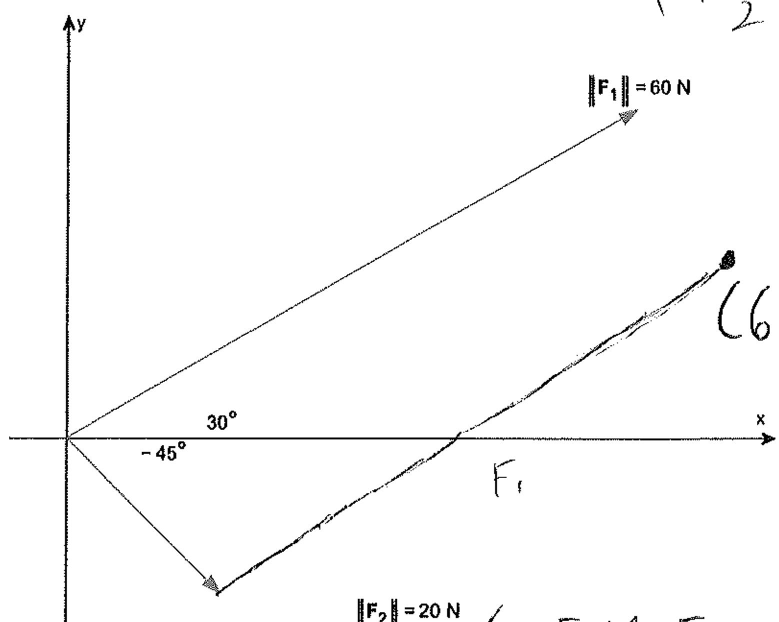
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$

$\frac{60}{1} \times \frac{\sqrt{3}}{2}$



(66.10365985, 37.81938861)

$\left(\frac{20\sqrt{2}+60\sqrt{3}}{2}, 30-\frac{20\sqrt{2}}{2}\right)$

Name ______________________ ID# ______________________

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

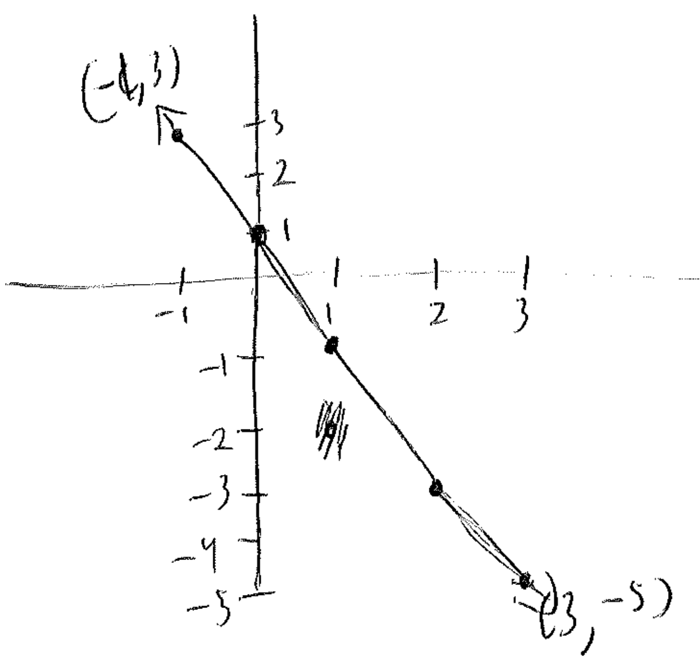
$4-(-1)^2$ $\quad 4-(3)^2$ $\quad \dfrac{(-5)-(3)}{(3)-(-1)} = \dfrac{-8}{4} = \boxed{-2}$

$4-1$ $\quad 4-9$

$(-1,3)$ $\quad (3,-5)$

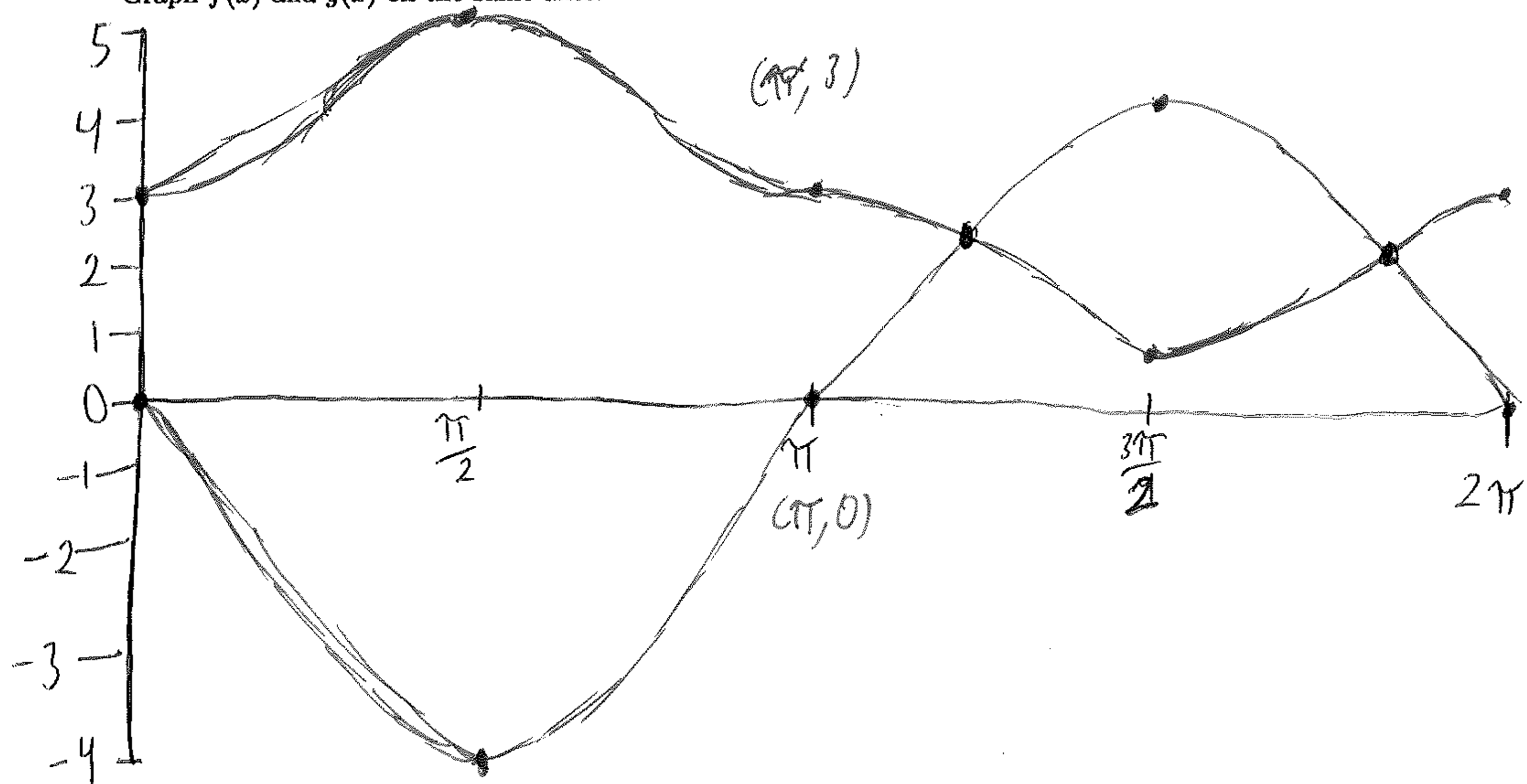
b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$y = -2x+1$ contains both



3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



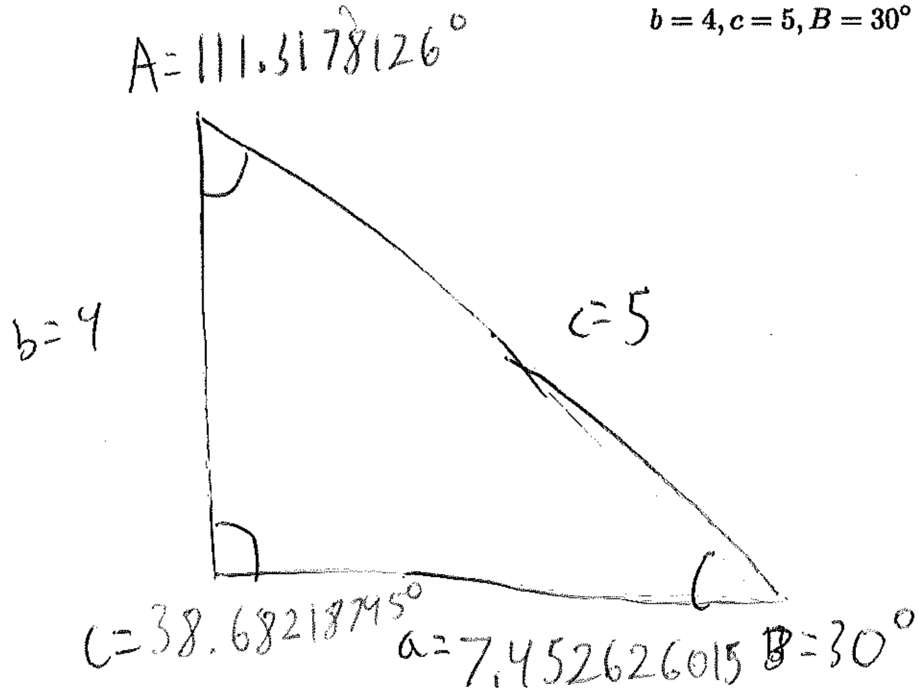
Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$\frac{7\pi}{6}, \frac{5\pi}{3}$

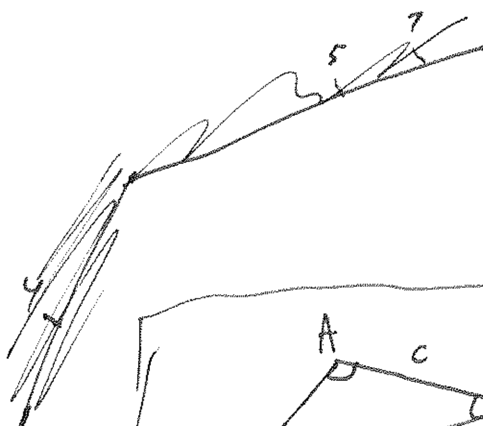
Name ______________________________ ID# ______________________________

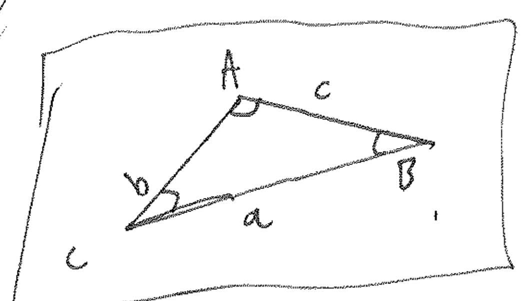
4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$b = 4, c = 5, B = 30°$



A=111.318°
C=38.682°
a=7.453

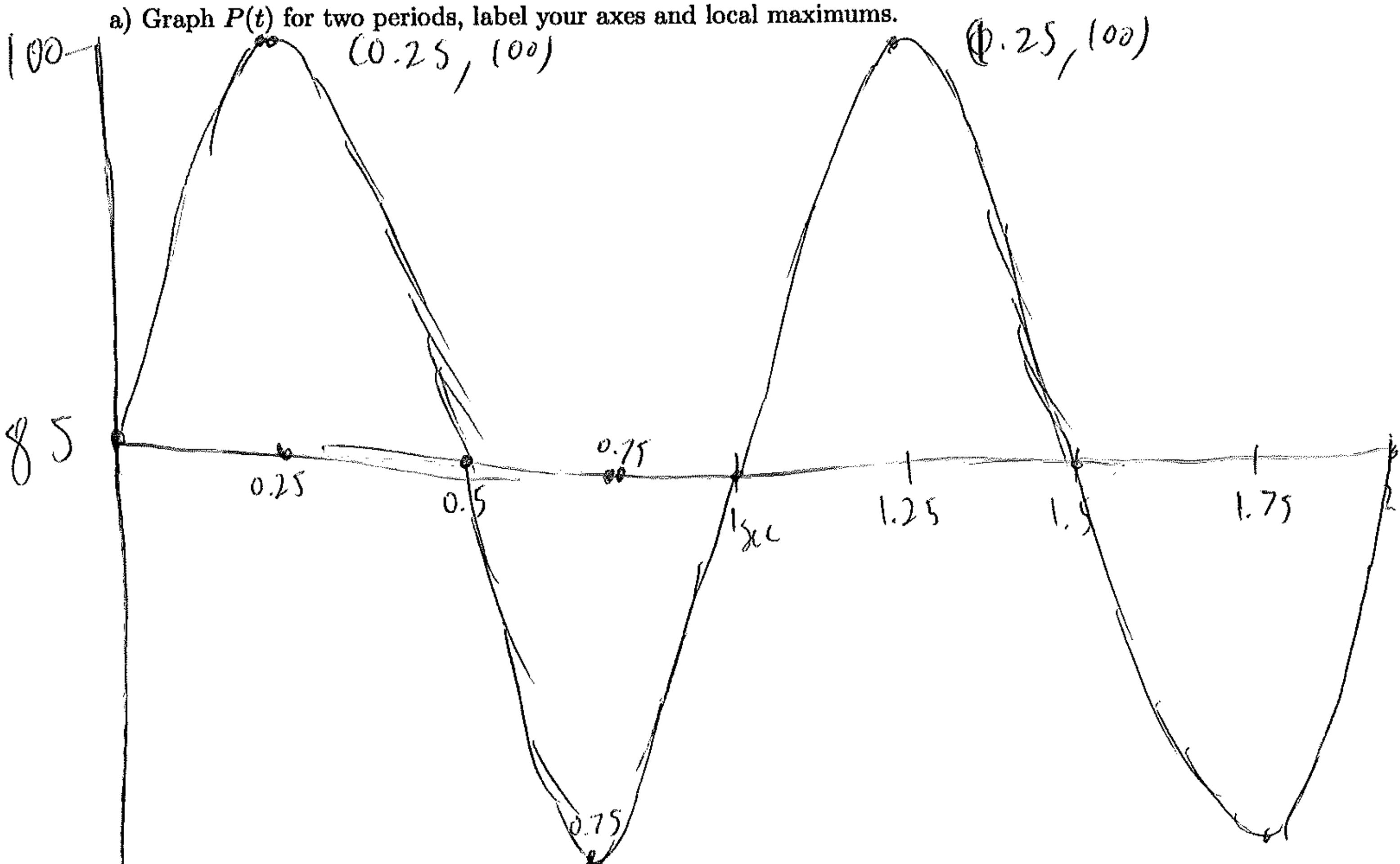




Measured too!!

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.



70

Max b) The systolic pressure is 100

Min c) The diastolic pressure is 70

d) The heart rate is 60 beats per minute

$$\frac{2\pi}{1} \times \frac{1}{2\pi} = \frac{2\pi}{2\pi} = 1 \text{ period}$$

Name [redacted] __________ **ID#** __________

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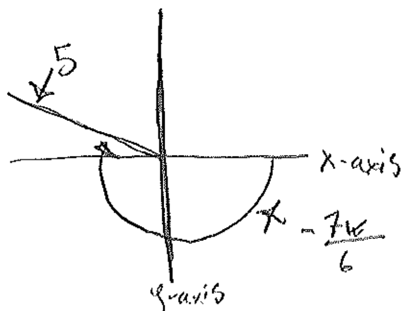
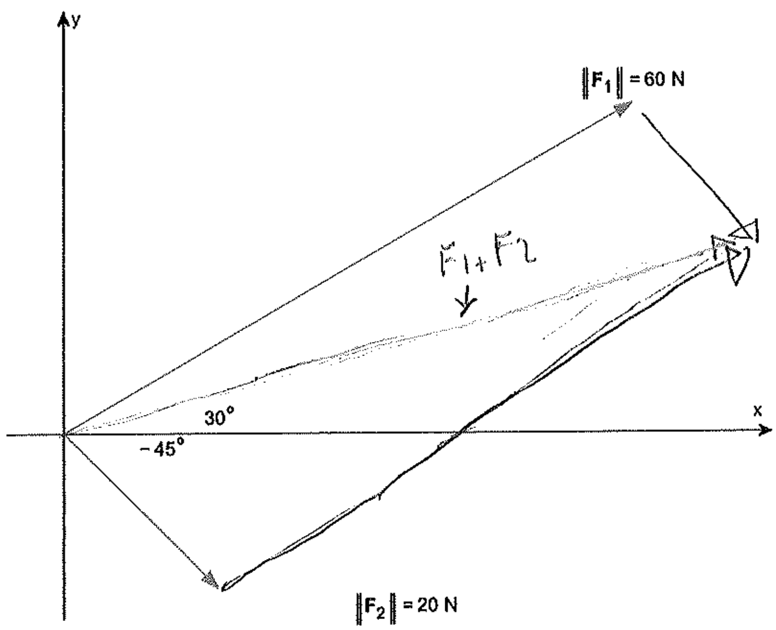
Total		/30

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$



$F_1 = (1 \cdot 60) = 60$

$\rightarrow \frac{\sqrt{3}}{2} \cdot \frac{60}{1}$

$\rightarrow 30\sqrt{3}i$

$\rightarrow \frac{1}{2} \cdot \frac{60}{1} = 30$

$\rightarrow 30\sqrt{3}i + 30j$

$F_1 = 30° = \frac{\pi}{6} \rightarrow$ $\frac{1}{2} = \|60\|$, $\frac{\sqrt{3}}{2}$

$F_2 = 45° = \frac{\pi}{4} =$ $\frac{\sqrt{2}}{2}$, 1, $\frac{\sqrt{2}}{2} = \|20\|$

$F_2 = (1 \cdot 20) = 20$

$\rightarrow \frac{\sqrt{2}}{2} \cdot \frac{60}{1} = 30\sqrt{2}i$

$\rightarrow \frac{\sqrt{2}}{2} \cdot \frac{60}{1} = -30\sqrt{2}j$

$F_1 + F_2 = (30\sqrt{3} + 30\sqrt{2})i + (30 - 30\sqrt{2})$

$\rightarrow F_1 + F_2 = 30\sqrt{3} + 30\sqrt{2}i - 30\sqrt{2}j$

Nam ______________________ **ID#** ______________________

2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

$f1 = g(-1) = 4-(-1)^2$

$\to g(-1) = 3$

$f2 = g(3) = 4-(3)^2$

$\to g(3) = 4-9$

$\to g(3) = -5$

$m = \dfrac{f(b)-f(a)}{b-a}$

$\to \dfrac{-5-3}{3+1}$

$\to \dfrac{-8}{4}$

$\to -2$

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$y - y1 = m(x - x1)$

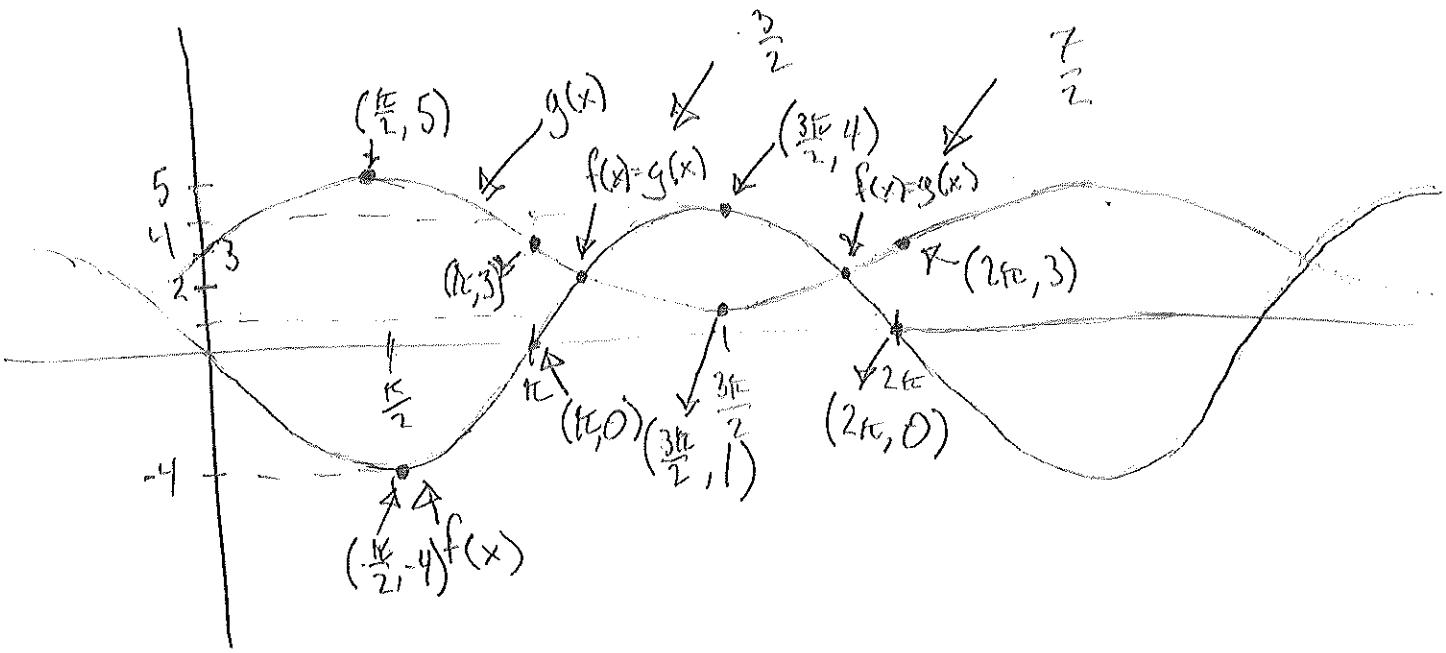
$\to y-3 = -2(x-(-1))$

$\to y-3 = -2x-2$
$\quad {}_{+3} \qquad\qquad {}_{+3}$

$\to y = -2x+1$

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$f(x) = g(x)$

$\to -4\sin(x) = 2\sin(x) + 3$

$\to -4\sin(x) - 3 = 2\sin(x)$

$\to \dfrac{-4\sin(x) - 3}{2} = \sin(x)$

$\to \sin^{-1}\left(\dfrac{-4\sin(x) - 3}{2}\right)$

$\to \dfrac{-4-3}{2}$

$\to \left|\dfrac{-7}{2}\right| \longrightarrow \dfrac{7}{2}, \dfrac{3}{2}$

$\to -4\sin(x) = 2\sin(x) + 3$

$\to \dfrac{2\sin(x) + 3}{-4} = \sin(x)$

$\to x = \sin^{-1}\left(\dfrac{2\sin(x) + 3}{-4}\right)$

$\to x = \dfrac{2+3}{-4}$

$\to \dfrac{6}{-4}$

$\to \left|-\dfrac{3}{2}\right|$

Name ______________ ID# ______________

4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

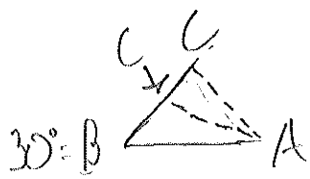
$b = 4, c = 5, B = 30°$

$$\frac{\sin(30°)}{4} = \frac{\sin(C)}{5}$$

$$\underset{C2}{\underset{\downarrow}{180-38.68}} = 141.32 + \underset{B}{\underset{\downarrow}{30}} = 171.32 \checkmark$$

$\rightarrow 5\sin(30°) = 4\sin(C)$

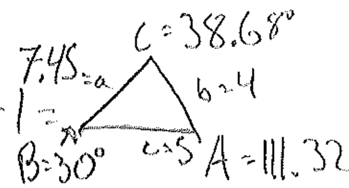
$\rightarrow \frac{5\sin(30°)}{4} = \sin(C)$



$\rightarrow \sin^{-1}\left(\frac{5\sin(30°)}{4}\right) = C$

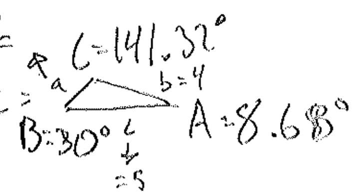
$\rightarrow C_1 \approx 38.6821$ or $\approx 38.68°$

triangle #1 =



$\rightarrow C2 \approx 141.32°$

triangle #2 =



$\rightarrow A_1 = 180 - (30 + 38.68)$

$\rightarrow A_1 = 180 - 68.68$

$\rightarrow A_1 = 111.32°$

$$a2 = \frac{\sin(30°)}{4} = \frac{\sin(8.68°)}{a}$$

$\rightarrow A_2 = 180° - (141.32° + 30°)$

$$\rightarrow a2 = \frac{4\sin(8.68°)}{\sin(30°)}$$

$\rightarrow A_2 = 180° - 171.32°$

$\rightarrow A2 = 8.68°$

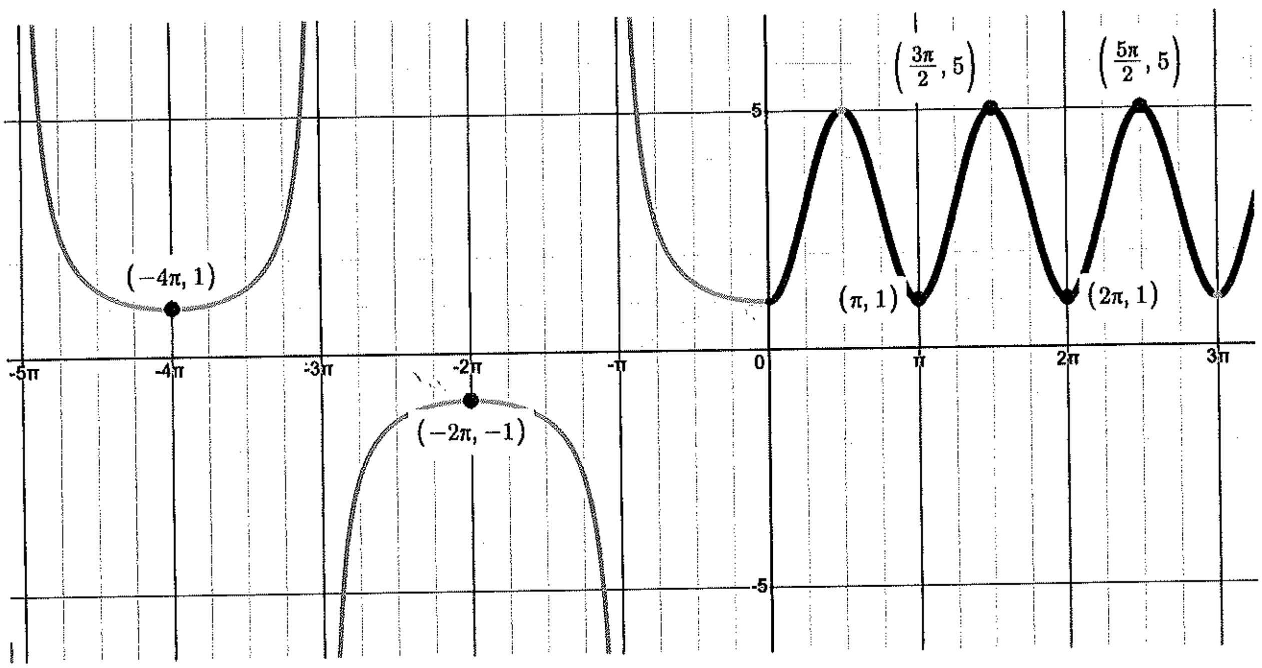
$\rightarrow a2 \approx 1.21$

$$a1 = \frac{\sin(30°)}{4} = \frac{\sin(111.32°)}{a}$$

$$\rightarrow a1 = \frac{4\sin(111.32°)}{\sin(30°)}$$

$\rightarrow a1 \approx 7.45$

5) Given the graph below find the function $f(x)$ and state the domain and range.



$$f(x) = \begin{cases} \text{when } x < 0 \to \sec\left(\frac{1}{2}x\right) \\ \text{when } x \geq 0 \to -2\cos(2x) + 3 \end{cases}$$

Range $= \mathbb{R}$

Domain $= (-\infty, 0)$ except any odd integer $\cdot \pi$
$\cup [0, \infty)$

black $f(x) =$
$\to a = \frac{1}{2}(5-1)$ $\quad b = \frac{2\pi}{\pi}$
$\to \frac{4}{2}$ $\quad \to 2$
$\to 2 = a$
$\to k = \frac{1}{2}(5+1)$
$\to \frac{6}{2}$
$\to k = 3$

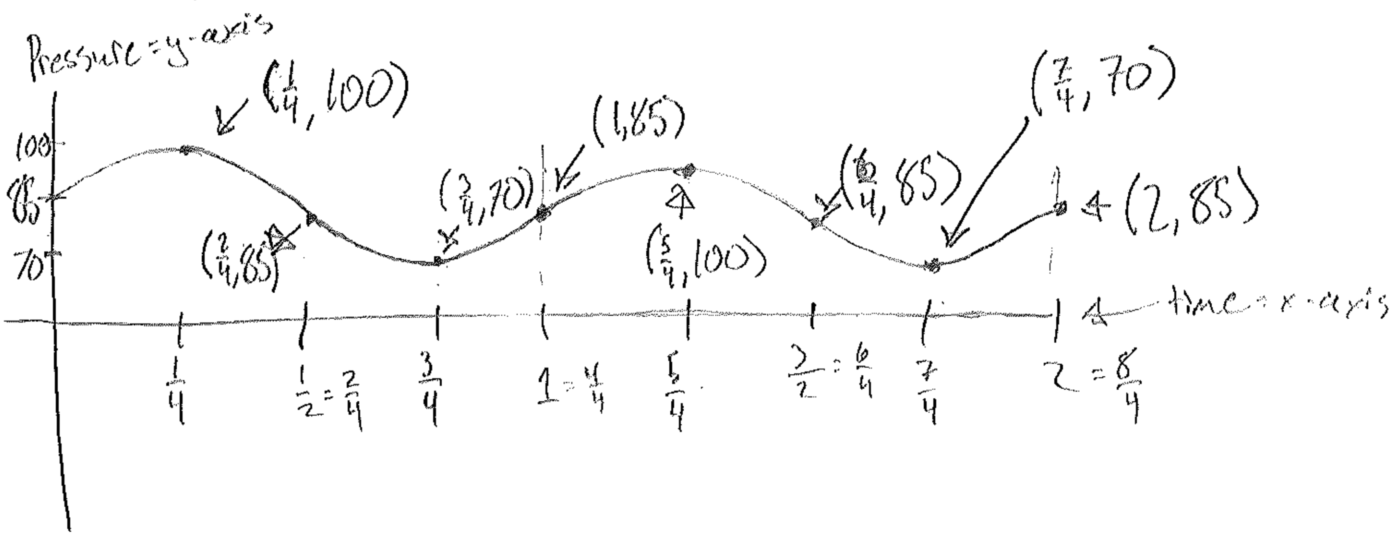
red $f(x) =$
$\to a = \frac{1}{2}(1+1)$
$\to a = 1$
$\to k = \frac{1}{2}(1+(-1))$
$\to k = \frac{0}{2} = 0$
$b = \frac{2\pi}{4\pi} = \frac{1}{2}$

Name ______________________ ID# ______________________

6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

Period $= \frac{2\pi}{2\pi} = 1$

a) Graph $P(t)$ for two periods, label your axes and local maximums.



heart rate: 1 sec for 1 beat (period = 1)

→ 60 sec in 1 min

→ $60 \div 1 = 60$

b) The systolic pressure is <u>100</u>

c) The diastolic pressure is <u>70</u>

d) The heart rate is <u>60</u> beats per minute